**UNIVERSIDAD NACIONAL MAYOR DE SAN MARCOS**

UNIVERSIDAD DEL PERÚ. DECANA DE AMÉRICA

**FACULTAD DE CIENCIAS ECONÓMICAS**

# “Inversores institucionales y el desarrollo bursátil en la Bolsa de Valores de Lima, 2014-2024”

Manual de pruebas estadísticas y econométricas del modelo.

**Asignatura:**

Seminario de Tesis II

**Docente:**

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**Lima – Perú**

**2024 – 2**

# Índice

El presente análisis tiene como objetivo mostrar los resultados de las pruebas econométricas del modelo Datos de Panel. El presente trabajo tiene 476 muestras de 4 instituciones: las 4 AFPs peruanas. Las siguientes variables son las utilizadas en el presente modelo:

D: Desarrollo Bursatil de la BVL

C: Capitalización de la BVL

AFP: Valor de cartera de cada una de las 4 AFPs

TRPM: Tasa de referencia de la Política Monetaria

INFLA: Variación porcentual de los precios

SPP: Cartera total del Sistema de Fondo de Pensiones (total de las 4 AFPs)

IGBVL: Índice General de la BVL

PENUSD: Tipo de cambio en soles de dólares

RETIRO: Monto retirado por Ley del fondo de pensiones.

# Análisis de Tendencia en Datos de Panel, un enfoque con efectos fijos y aleatorios

```
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Variable: D, Efectos Aleatorios - p-valor tendencia: 0.0000
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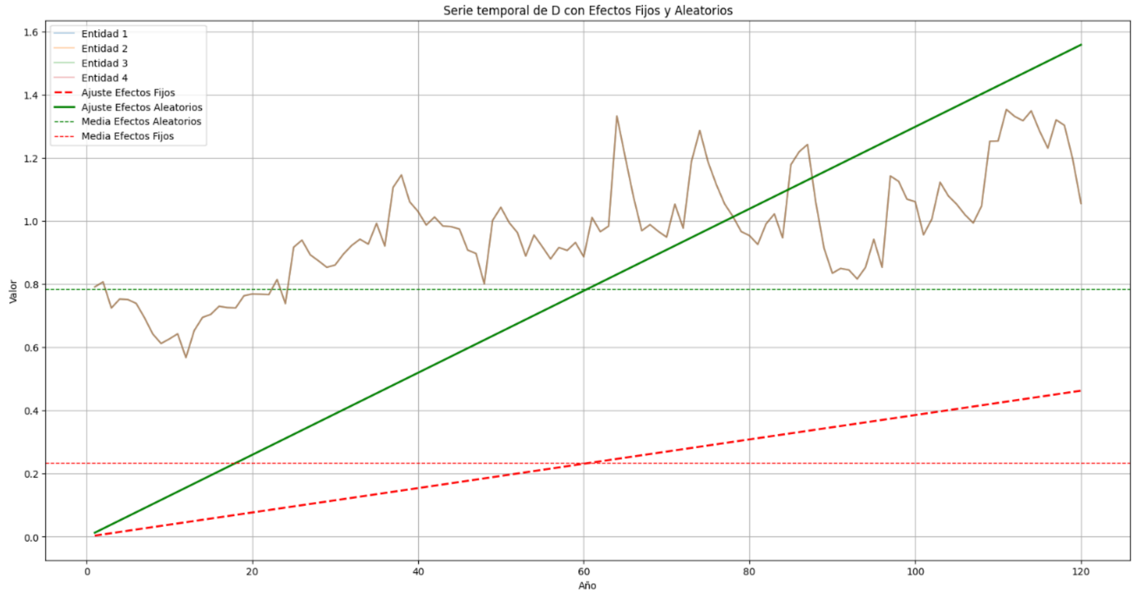


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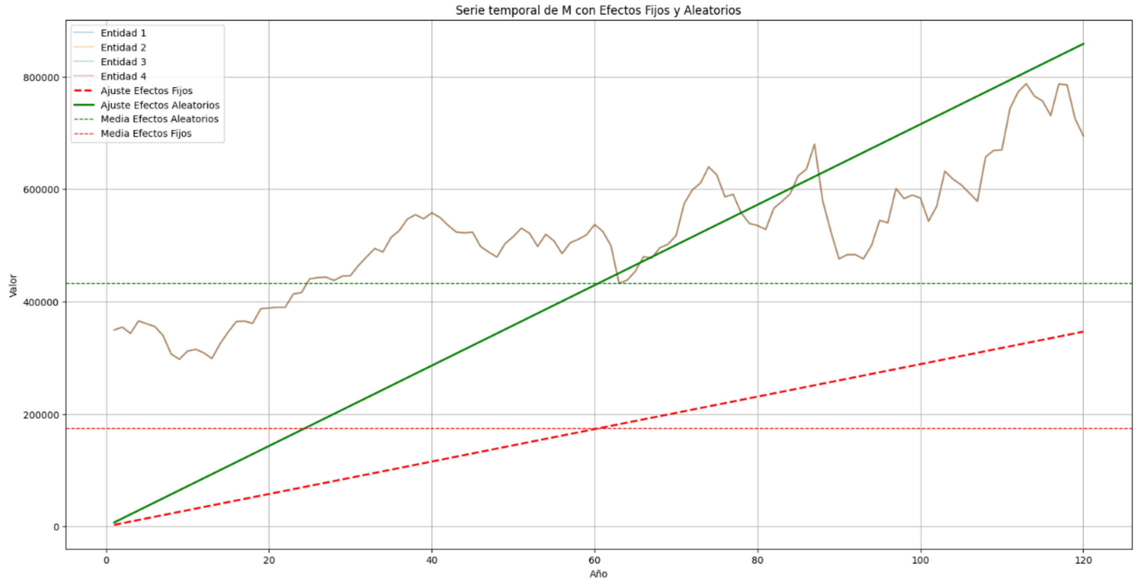


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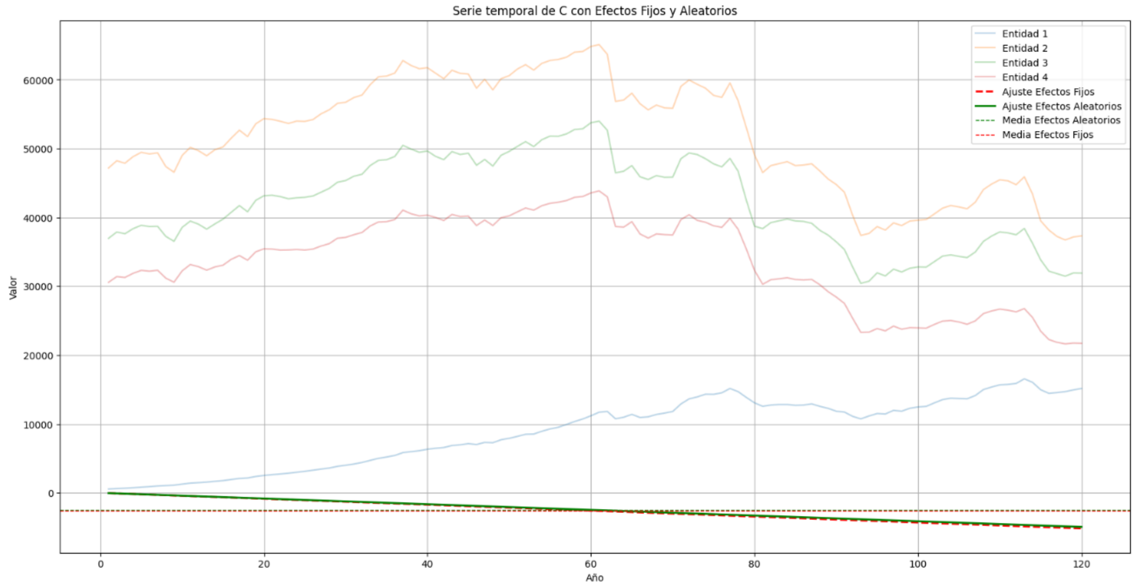


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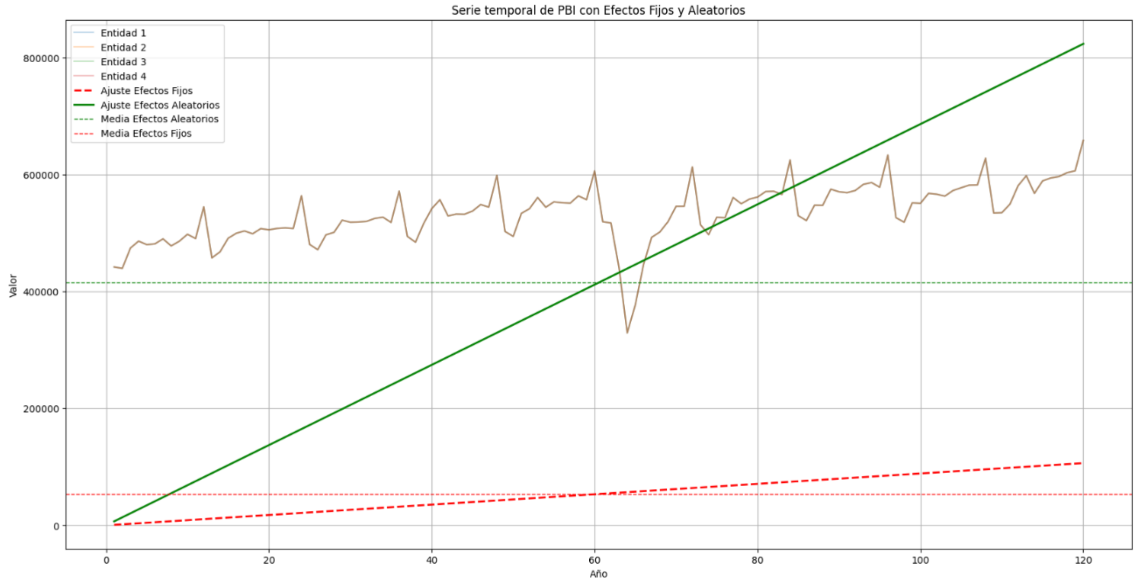


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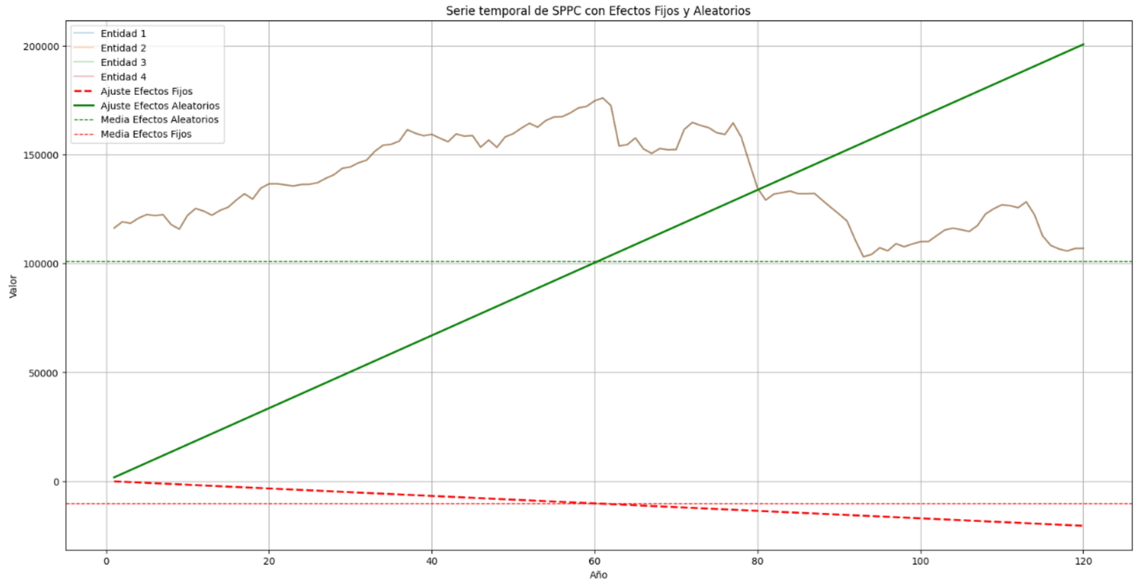


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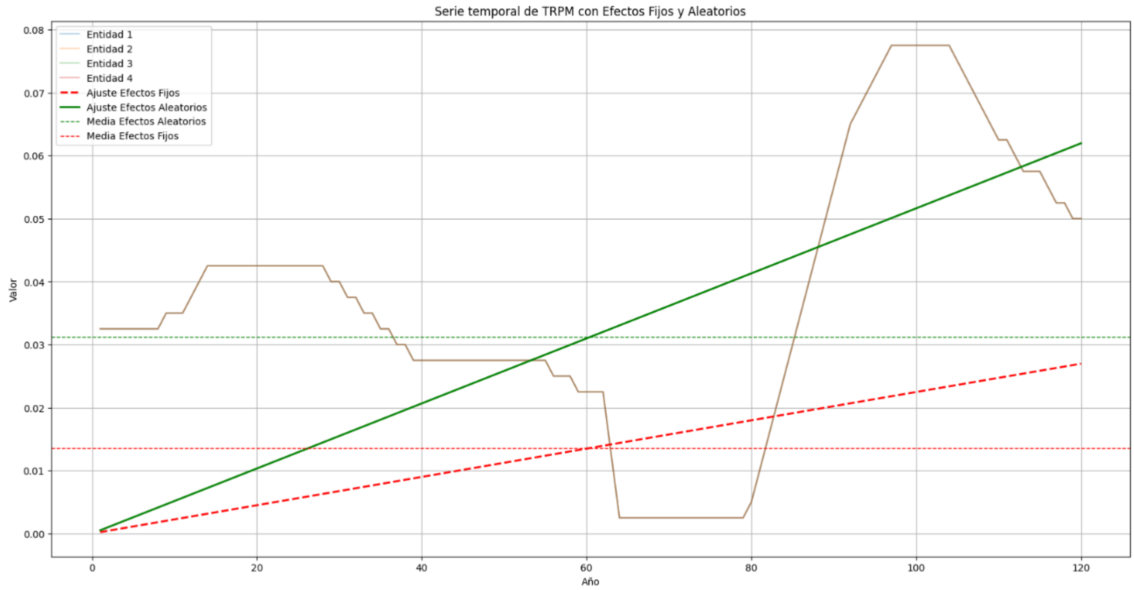


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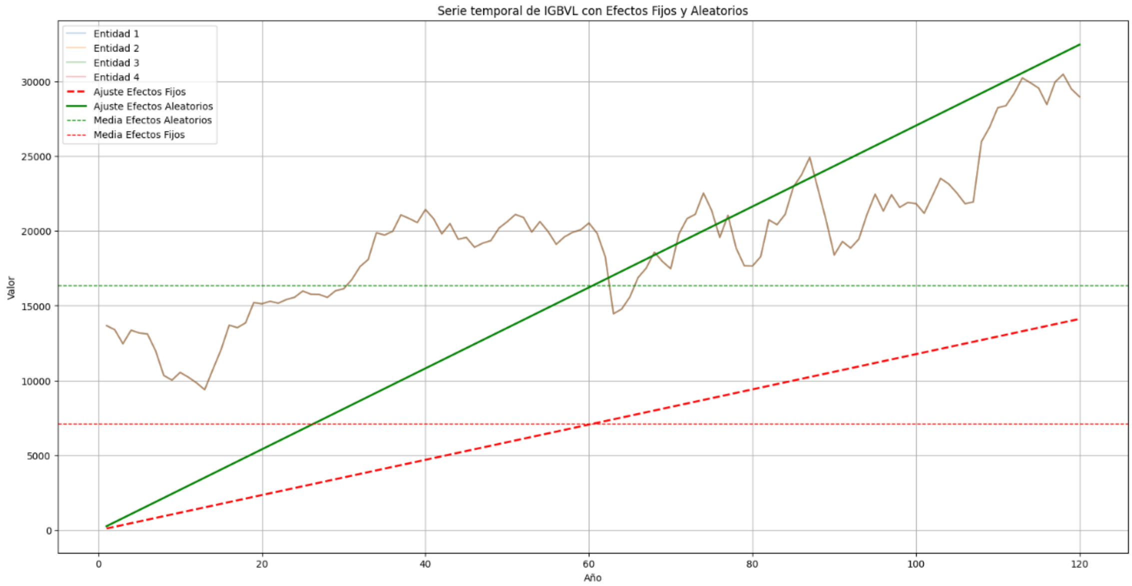


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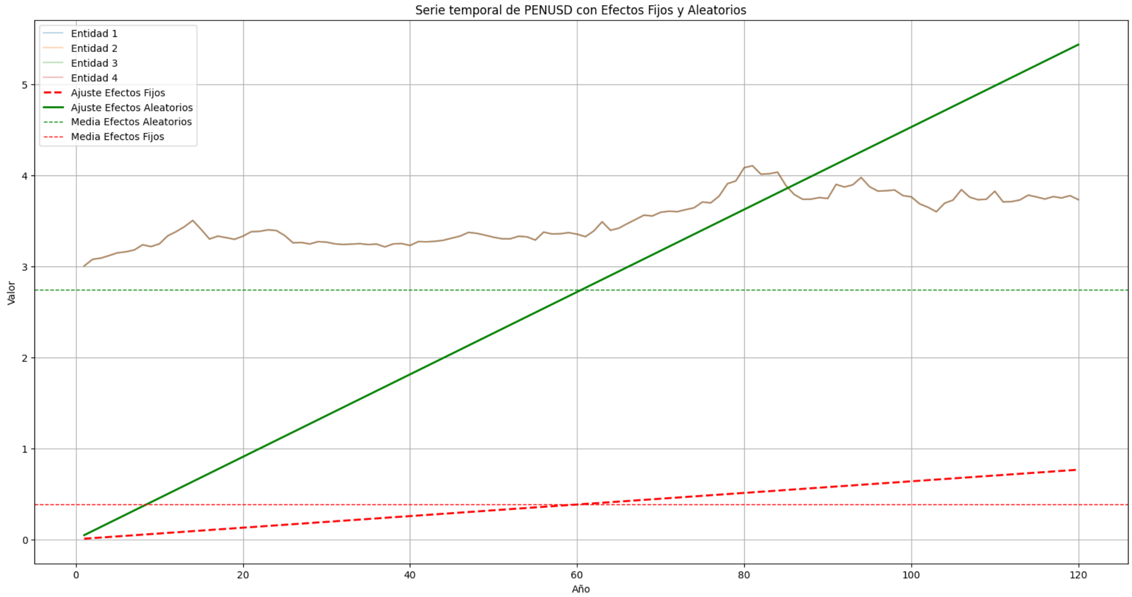


```
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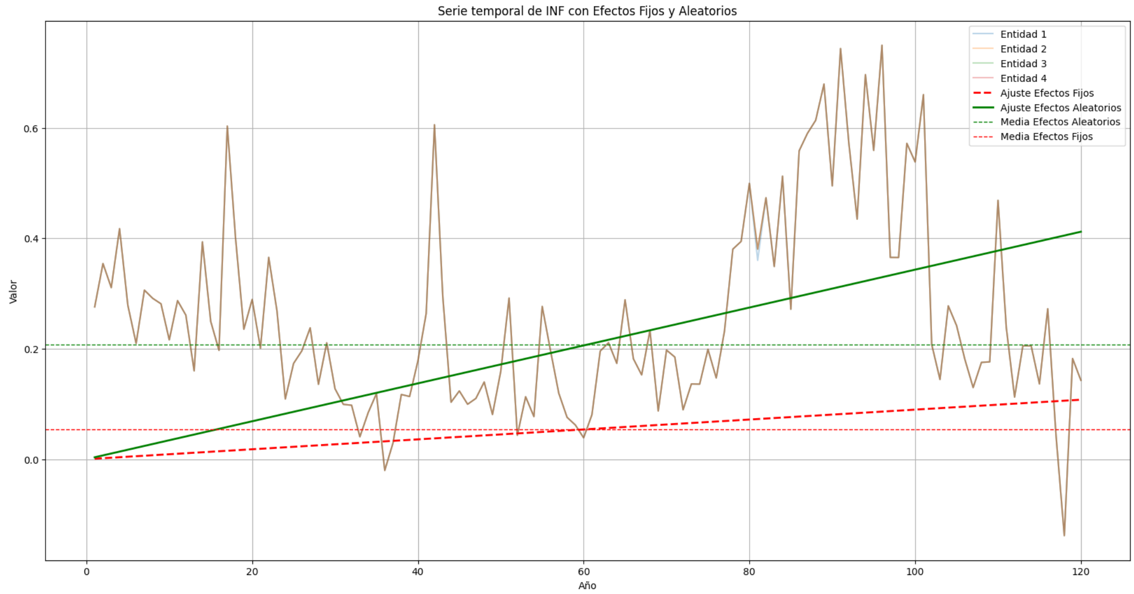
```

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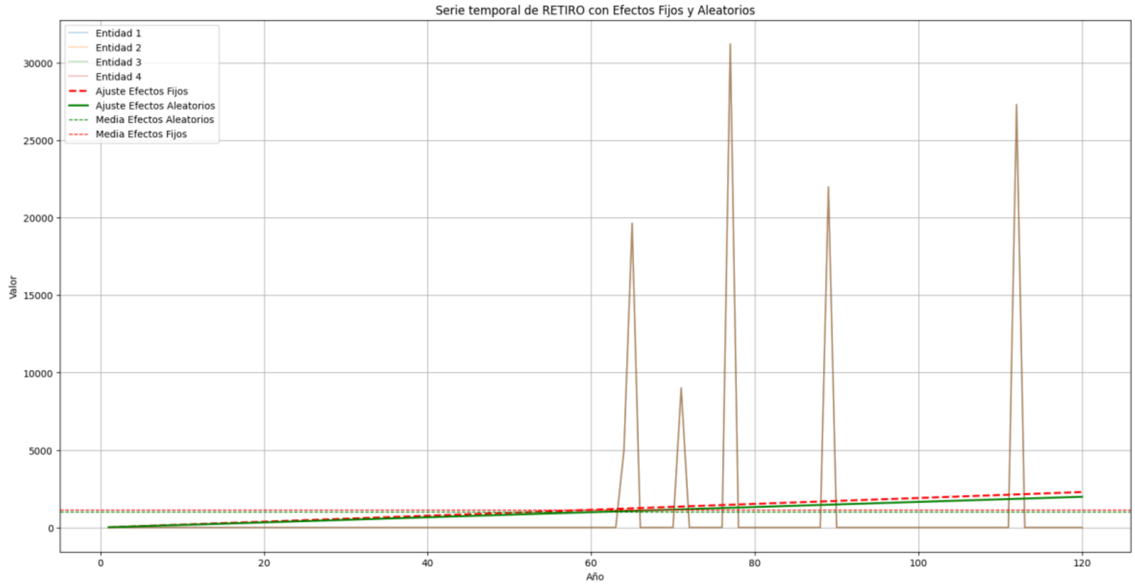


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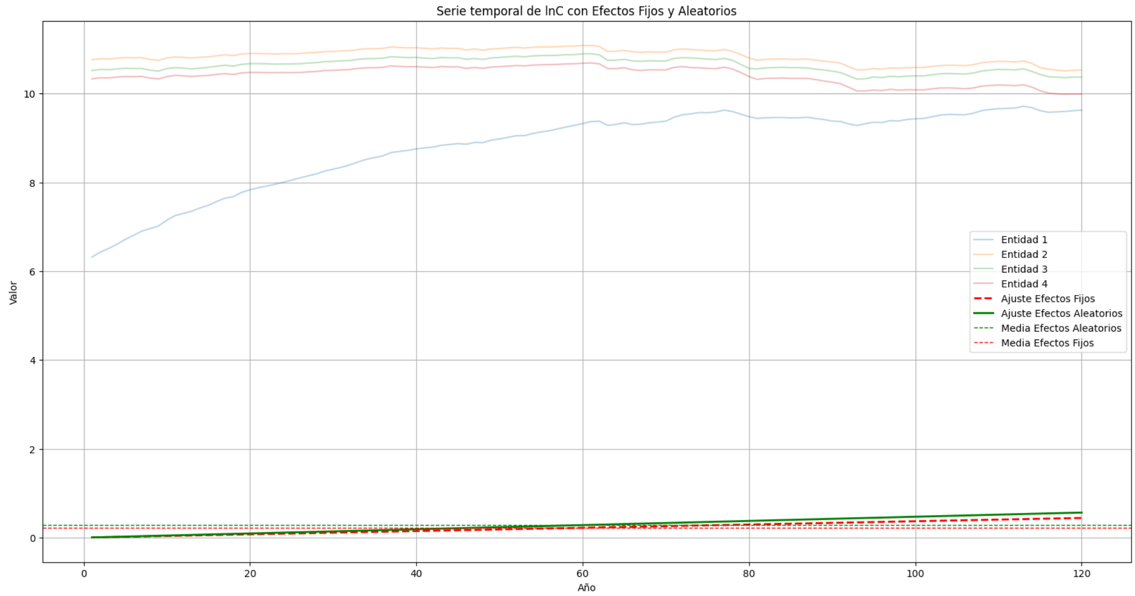


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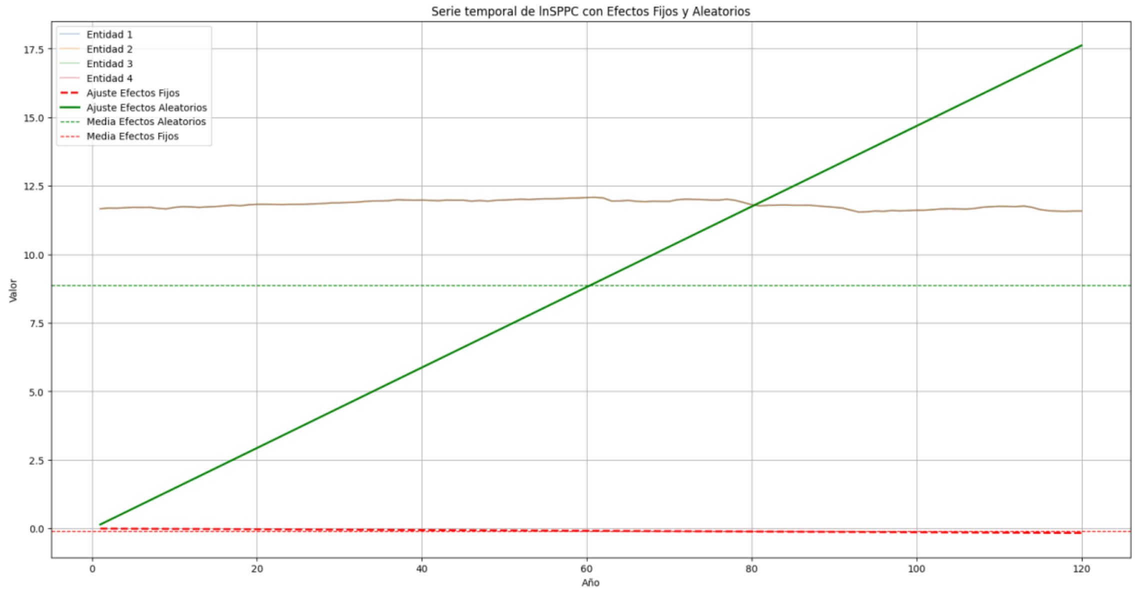


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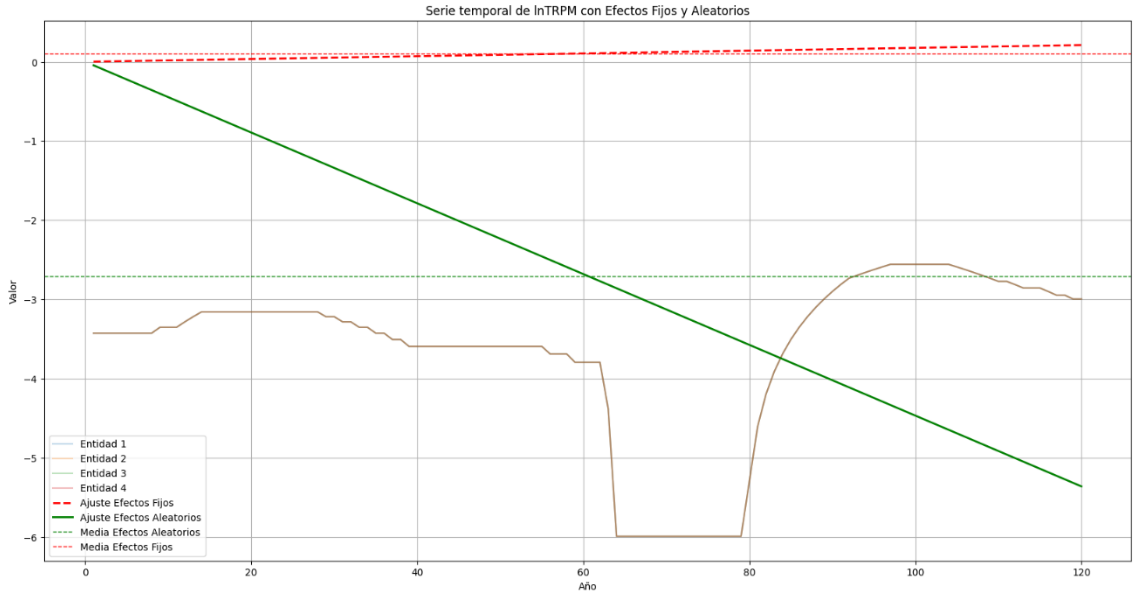


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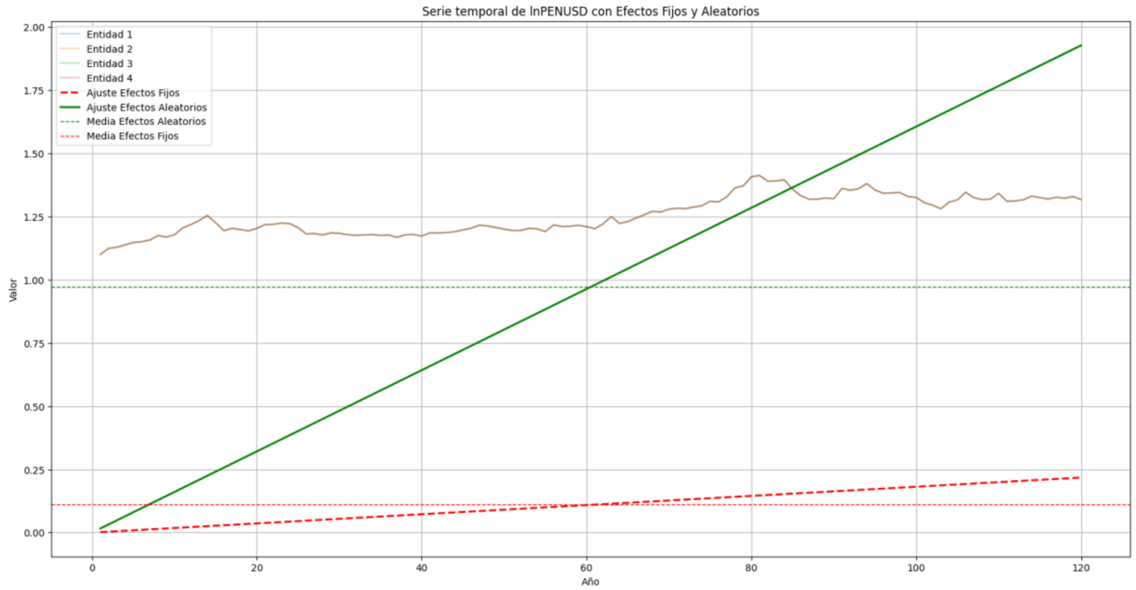


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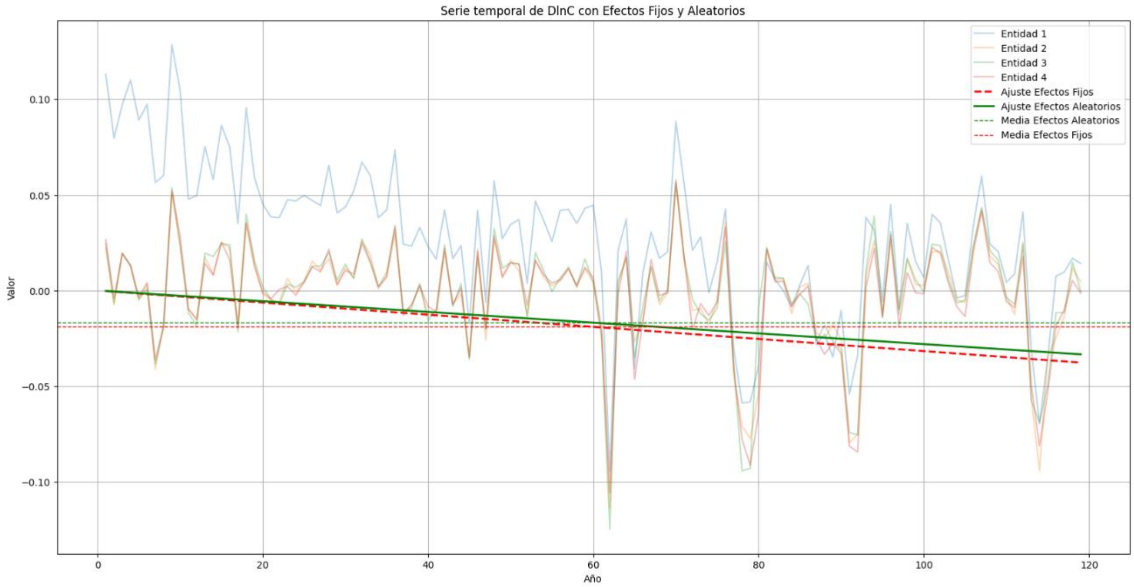


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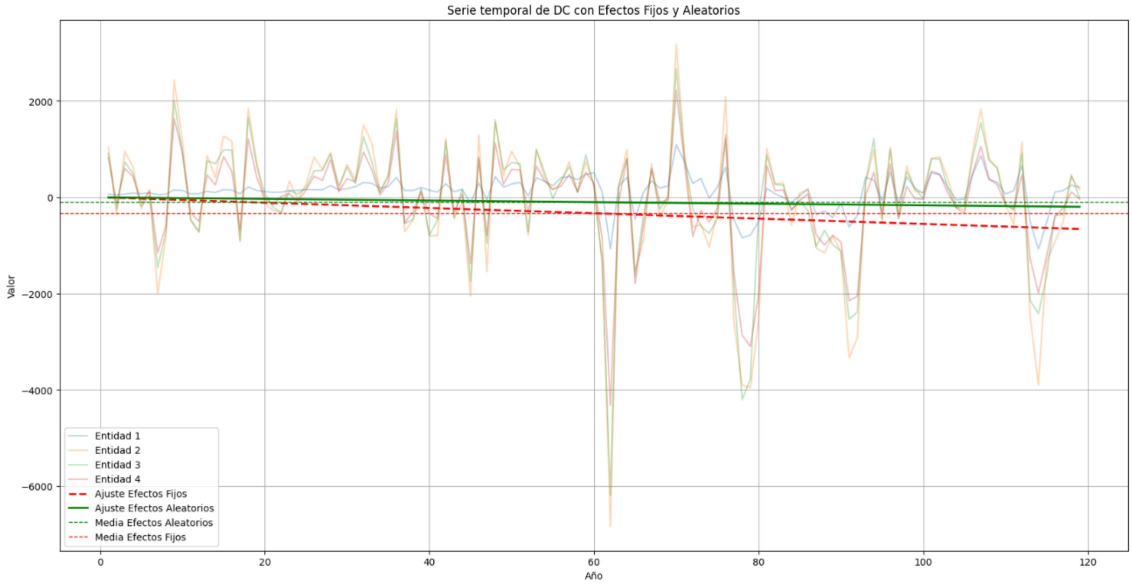


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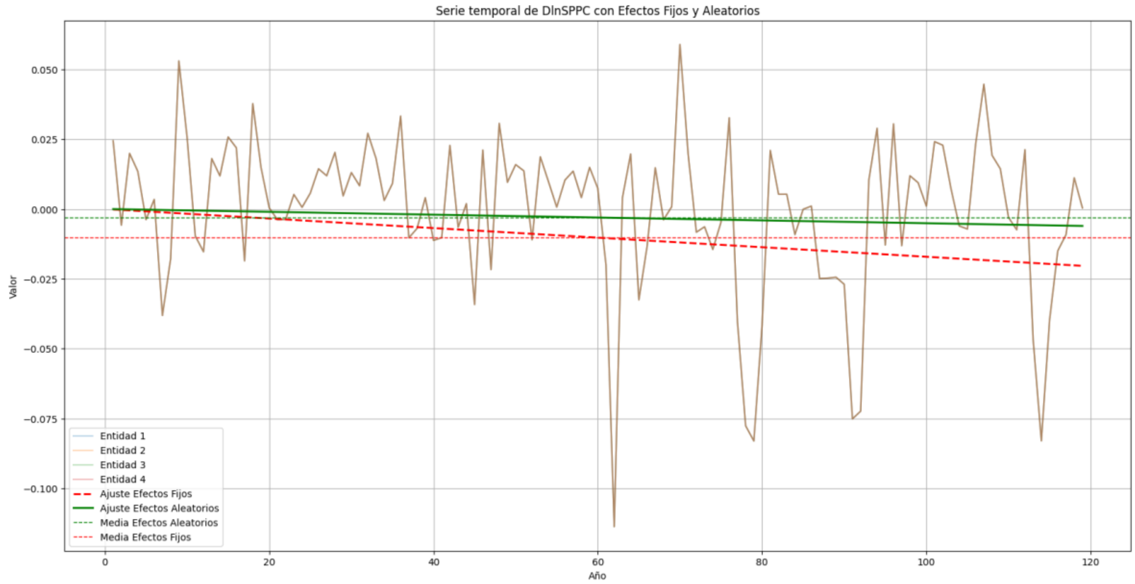


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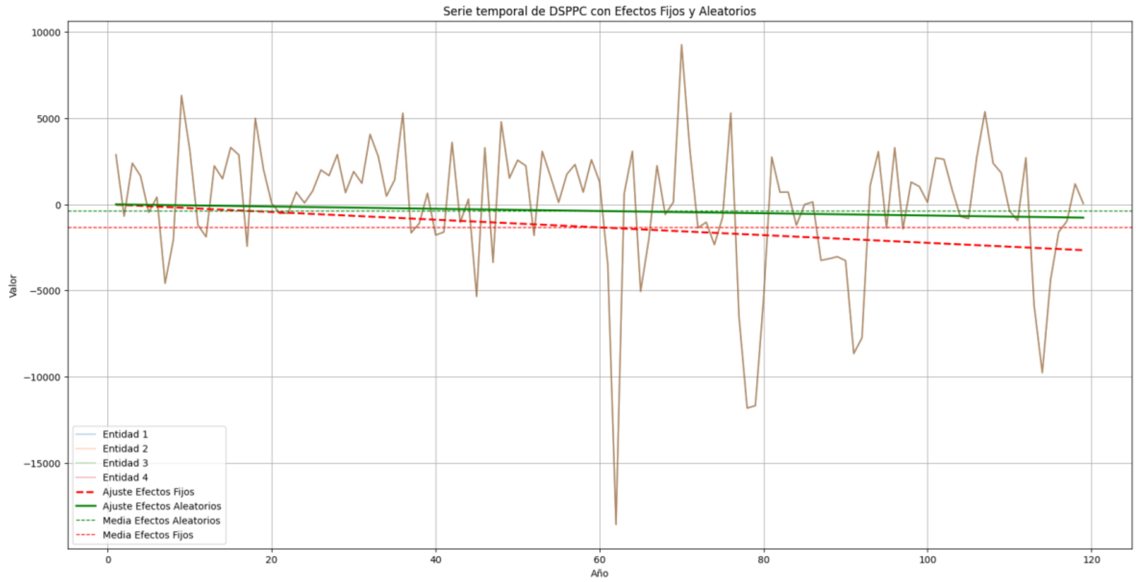


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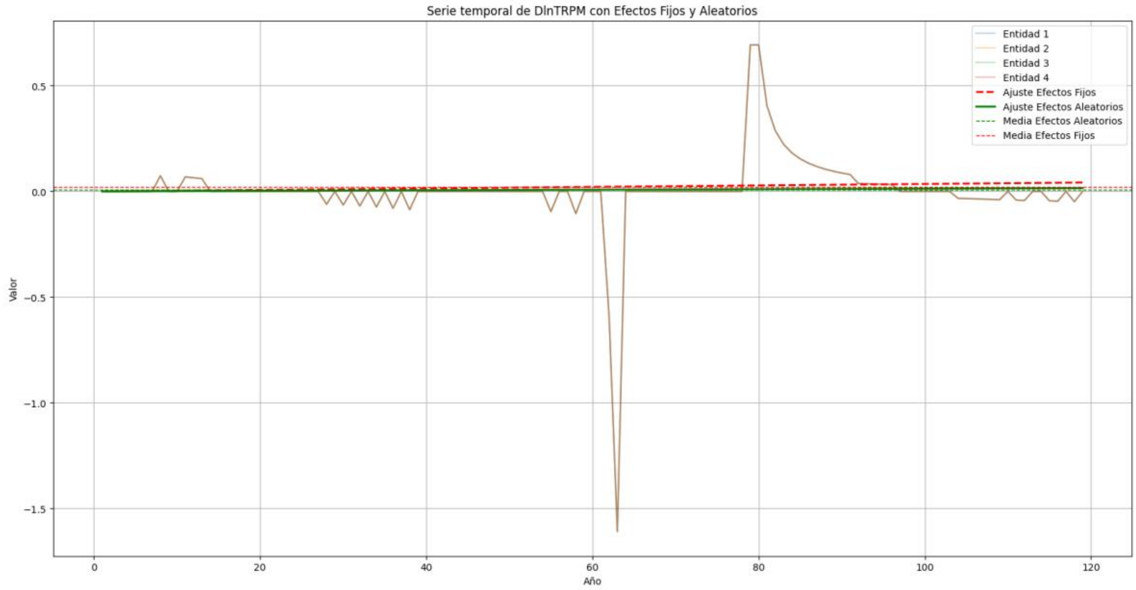


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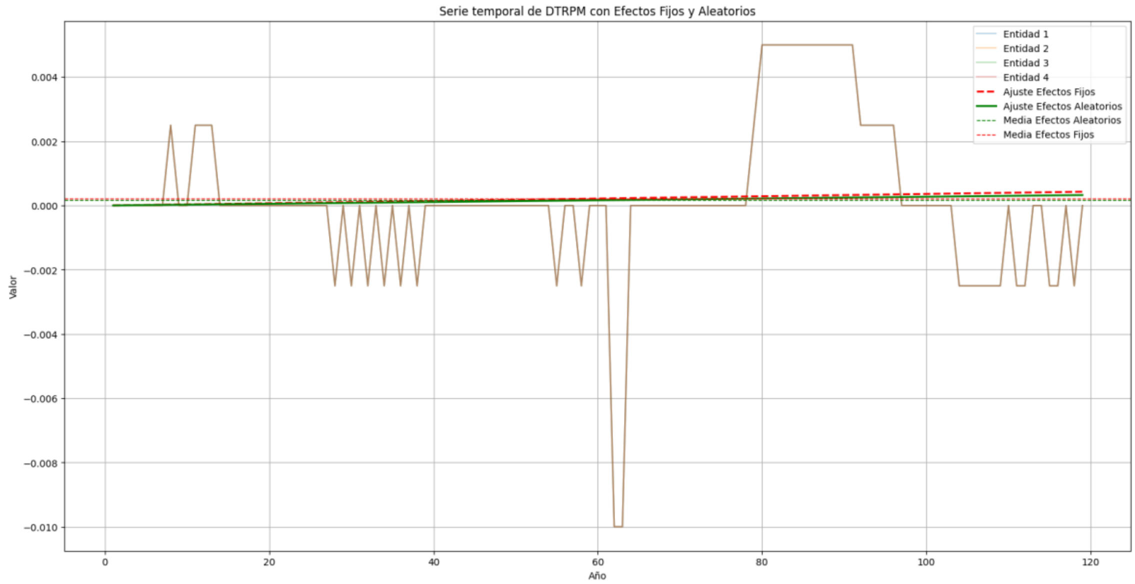


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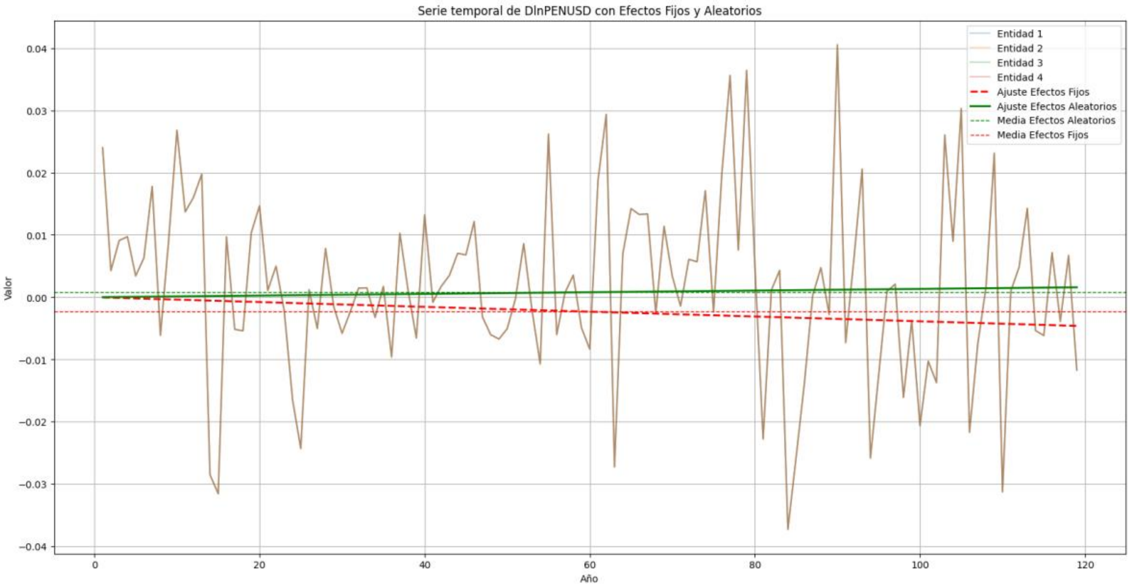


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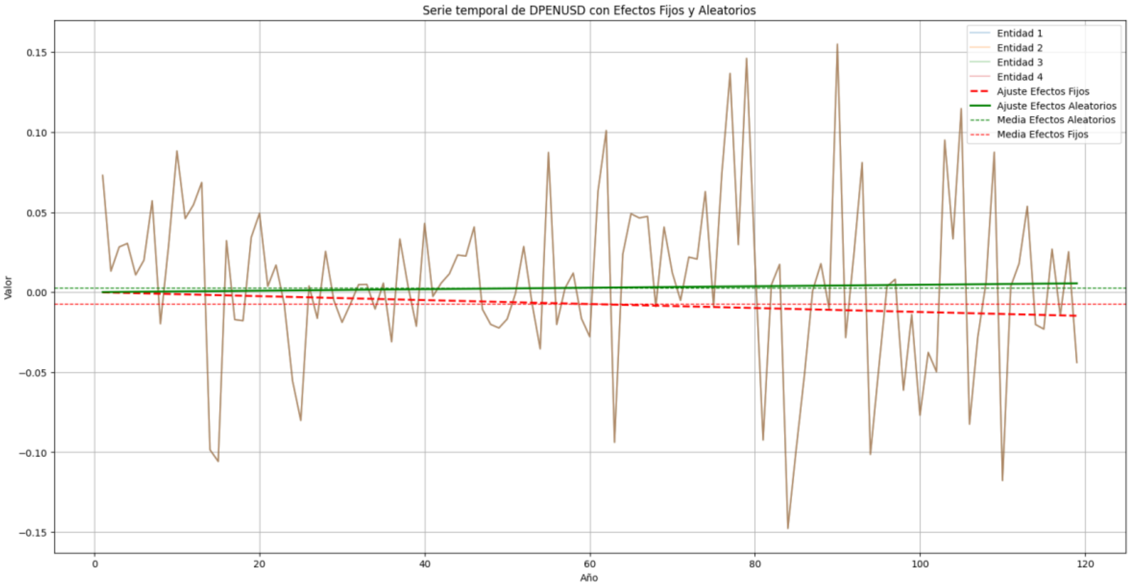


# Prueba de Estacionariedad Levin, Lin and Chu (LLC) e Im, Pesaran and Shin (IPS)

```
       panel variable:  AFP (strongly balanced)
        time variable:  Mes, 1 to 120
                delta:  1 unit

Levin-Lin-Chu unit-root test for D
----------------------------------
Ho: Panels contain unit roots            Number of panels  =      4
Ha: Panels are stationary                Number of periods =    120

AR parameter: Common                     Asymptotics: N/T -> 0
Panel means:  Included
Time trend:   Included

ADF regressions: 1 lag
LR variance:     Bartlett kernel, 15.00 lags average (chosen by LLC)
------------------------------------------------------------------------------
                    Statistic      p-value
------------------------------------------------------------------------------
 Unadjusted t        -8.0941
 Adjusted t*         -3.6641        0.0001
------------------------------------------------------------------------------

Im-Pesaran-Shin unit-root test for D
------------------------------------
Ho: All panels contain unit roots        Number of panels  =      4
Ha: Some panels are stationary           Number of periods =    120

AR parameter: Panel-specific             Asymptotics: T,N -> Infinity
Panel means:  Included                                  sequentially
Time trend:   Included

ADF regressions: No lags included
------------------------------------------------------------------------------
                                              Fixed-N exact critical values
                    Statistic      p-value         1%      5%      10%
------------------------------------------------------------------------------
 t-bar               -4.0162                     -2.990  -2.750  -2.620
 t-tilde-bar         -3.7949
 Z-t-tilde-bar       -5.5372        0.0000
------------------------------------------------------------------------------

Levin-Lin-Chu unit-root test for M
----------------------------------
Ho: Panels contain unit roots            Number of panels  =      4
Ha: Panels are stationary                Number of periods =    120

AR parameter: Common                     Asymptotics: N/T -> 0
Panel means:  Included
Time trend:   Included

ADF regressions: 1 lag
LR variance:     Bartlett kernel, 15.00 lags average (chosen by LLC)
------------------------------------------------------------------------------
                    Statistic      p-value
------------------------------------------------------------------------------
 Unadjusted t        -6.3947
 Adjusted t*         -2.4705        0.0067
------------------------------------------------------------------------------

Im-Pesaran-Shin unit-root test for M
------------------------------------
Ho: All panels contain unit roots        Number of panels  =      4
Ha: Some panels are stationary           Number of periods =    120

AR parameter: Panel-specific             Asymptotics: T,N -> Infinity
```

```
Panel means:  Included                                        sequentially
Time trend:   Included

ADF regressions: No lags included
------------------------------------------------------------------------------
                                              Fixed-N exact critical values
                    Statistic      p-value         1%      5%      10%
------------------------------------------------------------------------------
 t-bar               -2.5488                     -2.990  -2.750  -2.620
 t-tilde-bar         -2.5016
 Z-t-tilde-bar       -2.4096       0.0080
------------------------------------------------------------------------------

Levin-Lin-Chu unit-root test for C
----------------------------------
Ho: Panels contain unit roots              Number of panels  =      4
Ha: Panels are stationary                  Number of periods =    120

AR parameter: Common                       Asymptotics: N/T -> 0
Panel means:  Included
Time trend:   Included

ADF regressions: 1 lag
LR variance:     Bartlett kernel, 15.00 lags average (chosen by LLC)
------------------------------------------------------------------------------
                    Statistic      p-value
------------------------------------------------------------------------------
 Unadjusted t        -3.3972
 Adjusted t*         -1.0478       0.1474
------------------------------------------------------------------------------

Im-Pesaran-Shin unit-root test for C
------------------------------------
Ho: All panels contain unit roots          Number of panels  =      4
Ha: Some panels are stationary             Number of periods =    120

AR parameter: Panel-specific               Asymptotics: T,N -> Infinity
Panel means:  Included                                        sequentially
Time trend:   Included

ADF regressions: No lags included
------------------------------------------------------------------------------
                                              Fixed-N exact critical values
                    Statistic      p-value         1%      5%      10%
------------------------------------------------------------------------------
 t-bar               -1.3576                     -2.990  -2.750  -2.620
 t-tilde-bar         -1.3313
 Z-t-tilde-bar        0.4202       0.6628
------------------------------------------------------------------------------


Levin-Lin-Chu unit-root test for PBI
------------------------------------
Ho: Panels contain unit roots              Number of panels  =      4
Ha: Panels are stationary                  Number of periods =    120

AR parameter: Common                       Asymptotics: N/T -> 0
Panel means:  Included
Time trend:   Included

ADF regressions: 1 lag
LR variance:     Bartlett kernel, 15.00 lags average (chosen by LLC)
------------------------------------------------------------------------------
                    Statistic      p-value
------------------------------------------------------------------------------
 Unadjusted t       -10.0968
 Adjusted t*         -6.1516       0.0000
------------------------------------------------------------------------------
```

```
Im-Pesaran-Shin unit-root test for PBI
--------------------------------------
Ho: All panels contain unit roots           Number of panels  =      4
Ha: Some panels are stationary              Number of periods =    120

AR parameter: Panel-specific                Asymptotics: T,N -> Infinity
Panel means:  Included                                        sequentially
Time trend:   Included

ADF regressions: No lags included
------------------------------------------------------------------------------
                                              Fixed-N exact critical values
                 Statistic      p-value         1%      5%      10%
------------------------------------------------------------------------------
 t-bar             -5.6236                    -2.990  -2.750  -2.620
 t-tilde-bar       -5.0273
 Z-t-tilde-bar     -8.5174       0.0000
------------------------------------------------------------------------------


Levin-Lin-Chu unit-root test for SPPC
-------------------------------------
Ho: Panels contain unit roots               Number of panels  =      4
Ha: Panels are stationary                   Number of periods =    120

AR parameter: Common                        Asymptotics: N/T -> 0
Panel means:  Included
Time trend:   Included

ADF regressions: 1 lag
LR variance:     Bartlett kernel, 15.00 lags average (chosen by LLC)
------------------------------------------------------------------------------
                 Statistic      p-value
------------------------------------------------------------------------------
 Unadjusted t      -3.3309
 Adjusted t*       -1.2571       0.1044
------------------------------------------------------------------------------


Im-Pesaran-Shin unit-root test for SPPC
---------------------------------------
Ho: All panels contain unit roots           Number of panels  =      4
Ha: Some panels are stationary              Number of periods =    120

AR parameter: Panel-specific                Asymptotics: T,N -> Infinity
Panel means:  Included                                        sequentially
Time trend:   Included

ADF regressions: No lags included
------------------------------------------------------------------------------
                                              Fixed-N exact critical values
                 Statistic      p-value         1%      5%      10%
------------------------------------------------------------------------------
 t-bar             -1.4121                    -2.990  -2.750  -2.620
 t-tilde-bar       -1.3826
 Z-t-tilde-bar      0.2962       0.6164
------------------------------------------------------------------------------


Levin-Lin-Chu unit-root test for TRPM
-------------------------------------
Ho: Panels contain unit roots               Number of panels  =      4
Ha: Panels are stationary                   Number of periods =    120

AR parameter: Common                        Asymptotics: N/T -> 0
Panel means:  Included
Time trend:   Included

ADF regressions: 1 lag
LR variance:     Bartlett kernel, 15.00 lags average (chosen by LLC)
------------------------------------------------------------------------------
```

```
                    Statistic      p-value
------------------------------------------------------------------------------
 Unadjusted t        -3.2002
 Adjusted t*          0.9622       0.8320
------------------------------------------------------------------------------


Im-Pesaran-Shin unit-root test for TRPM
---------------------------------------
Ho: All panels contain unit roots           Number of panels  =      4
Ha: Some panels are stationary              Number of periods =    120

AR parameter: Panel-specific                Asymptotics: T,N -> Infinity
Panel means:  Included                                        sequentially
Time trend:   Included

ADF regressions: No lags included
------------------------------------------------------------------------------
                                              Fixed-N exact critical values
                    Statistic      p-value         1%      5%      10%
------------------------------------------------------------------------------
 t-bar               -0.7161                     -2.990  -2.750  -2.620
 t-tilde-bar         -0.7197
 Z-t-tilde-bar        1.8992       0.9712
------------------------------------------------------------------------------


Levin-Lin-Chu unit-root test for IGBVL
--------------------------------------
Ho: Panels contain unit roots               Number of panels  =      4
Ha: Panels are stationary                   Number of periods =    120

AR parameter: Common                        Asymptotics: N/T -> 0
Panel means:  Included
Time trend:   Included

ADF regressions: 1 lag
LR variance:     Bartlett kernel, 15.00 lags average (chosen by LLC)
------------------------------------------------------------------------------
                    Statistic      p-value
------------------------------------------------------------------------------
 Unadjusted t        -5.7784
 Adjusted t*         -2.2624       0.0118
------------------------------------------------------------------------------



Im-Pesaran-Shin unit-root test for IGBVL
----------------------------------------
Ho: All panels contain unit roots           Number of panels  =      4
Ha: Some panels are stationary              Number of periods =    120

AR parameter: Panel-specific                Asymptotics: T,N -> Infinity
Panel means:  Included                                        sequentially
Time trend:   Included

ADF regressions: No lags included
------------------------------------------------------------------------------
                                              Fixed-N exact critical values
                    Statistic      p-value         1%      5%      10%
------------------------------------------------------------------------------
 t-bar               -2.2809                     -2.990  -2.750  -2.620
 t-tilde-bar         -2.2475
 Z-t-tilde-bar       -1.7952       0.0363
------------------------------------------------------------------------------


Levin-Lin-Chu unit-root test for PENUSD
---------------------------------------
Ho: Panels contain unit roots               Number of panels  =      4
Ha: Panels are stationary                   Number of periods =    120
```

```
AR parameter: Common                     Asymptotics: N/T -> 0
Panel means:  Included
Time trend:   Included

ADF regressions: 1 lag
LR variance:     Bartlett kernel, 15.00 lags average (chosen by LLC)
------------------------------------------------------------------------------
                    Statistic      p-value
------------------------------------------------------------------------------
 Unadjusted t        -4.2002
 Adjusted t*          0.3734       0.6456
------------------------------------------------------------------------------


Im-Pesaran-Shin unit-root test for PENUSD
-----------------------------------------
Ho: All panels contain unit roots          Number of panels  =      4
Ha: Some panels are stationary             Number of periods =    120

AR parameter: Panel-specific               Asymptotics: T,N -> Infinity
Panel means:  Included                                       sequentially
Time trend:   Included

ADF regressions: No lags included
------------------------------------------------------------------------------
                                             Fixed-N exact critical values
                    Statistic      p-value         1%      5%      10%
------------------------------------------------------------------------------
 t-bar               -1.9582                    -2.990  -2.750  -2.620
 t-tilde-bar         -1.9367
 Z-t-tilde-bar       -1.0438       0.1483
------------------------------------------------------------------------------


Levin-Lin-Chu unit-root test for INF
------------------------------------
Ho: Panels contain unit roots              Number of panels  =      4
Ha: Panels are stationary                  Number of periods =    120

AR parameter: Common                       Asymptotics: N/T -> 0
Panel means:  Included
Time trend:   Included

ADF regressions: 1 lag
LR variance:     Bartlett kernel, 15.00 lags average (chosen by LLC)
------------------------------------------------------------------------------
                    Statistic      p-value
------------------------------------------------------------------------------
 Unadjusted t        -6.8192
 Adjusted t*         -3.0934       0.0010
------------------------------------------------------------------------------


Im-Pesaran-Shin unit-root test for INF
--------------------------------------
Ho: All panels contain unit roots          Number of panels  =      4
Ha: Some panels are stationary             Number of periods =    120

AR parameter: Panel-specific               Asymptotics: T,N -> Infinity
Panel means:  Included                                       sequentially
Time trend:   Included

ADF regressions: No lags included
------------------------------------------------------------------------------
                                             Fixed-N exact critical values
                    Statistic      p-value         1%      5%      10%
------------------------------------------------------------------------------
 t-bar               -4.7574                    -2.990  -2.750  -2.620
 t-tilde-bar         -4.3889
 Z-t-tilde-bar       -6.9735       0.0000
```

```
-------------------------------------------------------------------------------


Levin-Lin-Chu unit-root test for RETIRO
---------------------------------------
Ho: Panels contain unit roots             Number of panels  =      4
Ha: Panels are stationary                 Number of periods =    120

AR parameter: Common                      Asymptotics: N/T -> 0
Panel means:  Included
Time trend:   Included

ADF regressions: 1 lag
LR variance:     Bartlett kernel, 15.00 lags average (chosen by LLC)
-------------------------------------------------------------------------------
                    Statistic      p-value
-------------------------------------------------------------------------------
 Unadjusted t       -16.5924
 Adjusted t*        -14.9535        0.0000
-------------------------------------------------------------------------------



Im-Pesaran-Shin unit-root test for RETIRO
-----------------------------------------
Ho: All panels contain unit roots         Number of panels  =      4
Ha: Some panels are stationary            Number of periods =    120

AR parameter: Panel-specific              Asymptotics: T,N -> Infinity
Panel means:  Included                                  sequentially
Time trend:   Included

ADF regressions: No lags included
-------------------------------------------------------------------------------
                                            Fixed-N exact critical values
                    Statistic      p-value        1%      5%      10%
-------------------------------------------------------------------------------
 t-bar              -11.0438                    -2.990  -2.750  -2.620
 t-tilde-bar         -7.7767
 Z-t-tilde-bar      -15.1659        0.0000
-------------------------------------------------------------------------------


Levin-Lin-Chu unit-root test for lnC
------------------------------------
Ho: Panels contain unit roots             Number of panels  =      4
Ha: Panels are stationary                 Number of periods =    120

AR parameter: Common                      Asymptotics: N/T -> 0
Panel means:  Included
Time trend:   Included

ADF regressions: 1 lag
LR variance:     Bartlett kernel, 15.00 lags average (chosen by LLC)
-------------------------------------------------------------------------------
                    Statistic      p-value
-------------------------------------------------------------------------------
 Unadjusted t        -4.3467
 Adjusted t*         -3.5582        0.0002
-------------------------------------------------------------------------------


Im-Pesaran-Shin unit-root test for lnC
--------------------------------------
Ho: All panels contain unit roots         Number of panels  =      4
Ha: Some panels are stationary            Number of periods =    120

AR parameter: Panel-specific              Asymptotics: T,N -> Infinity
Panel means:  Included                                  sequentially
Time trend:   Included
```

```
ADF regressions: No lags included
------------------------------------------------------------------------------
                                              Fixed-N exact critical values
                    Statistic      p-value        1%      5%      10%
------------------------------------------------------------------------------
 t-bar              -2.2146                    -2.990  -2.750  -2.620
 t-tilde-bar        -1.8626
 Z-t-tilde-bar      -0.8645       0.1937
------------------------------------------------------------------------------


Levin-Lin-Chu unit-root test for lnSPPC
---------------------------------------
Ho: Panels contain unit roots             Number of panels  =      4
Ha: Panels are stationary                 Number of periods =    120

AR parameter: Common                      Asymptotics: N/T -> 0
Panel means:  Included
Time trend:   Included

ADF regressions: 1 lag
LR variance:     Bartlett kernel, 15.00 lags average (chosen by LLC)
------------------------------------------------------------------------------
                    Statistic      p-value
------------------------------------------------------------------------------
 Unadjusted t       -3.3684
 Adjusted t*        -1.1936       0.1163
------------------------------------------------------------------------------


Im-Pesaran-Shin unit-root test for lnSPPC
-----------------------------------------
Ho: All panels contain unit roots         Number of panels  =      4
Ha: Some panels are stationary            Number of periods =    120

AR parameter: Panel-specific              Asymptotics: T,N -> Infinity
Panel means:  Included                                  sequentially
Time trend:   Included

ADF regressions: No lags included
------------------------------------------------------------------------------
                                              Fixed-N exact critical values
                    Statistic      p-value        1%      5%      10%
------------------------------------------------------------------------------
 t-bar              -1.4163                    -2.990  -2.750  -2.620
 t-tilde-bar        -1.3830
 Z-t-tilde-bar       0.2952       0.6161
------------------------------------------------------------------------------


Levin-Lin-Chu unit-root test for lnTRPM
---------------------------------------
Ho: Panels contain unit roots             Number of panels  =      4
Ha: Panels are stationary                 Number of periods =    120

AR parameter: Common                      Asymptotics: N/T -> 0
Panel means:  Included
Time trend:   Not included

ADF regressions: 1 lag
LR variance:     Bartlett kernel, 15.00 lags average (chosen by LLC)
------------------------------------------------------------------------------
                    Statistic      p-value
------------------------------------------------------------------------------
 Unadjusted t       -3.3291
 Adjusted t*         0.5722       0.7164
------------------------------------------------------------------------------


Im-Pesaran-Shin unit-root test for lnTRPM
```

```
-----------------------------------------
Ho: All panels contain unit roots           Number of panels  =      4
Ha: Some panels are stationary              Number of periods =    120

AR parameter: Panel-specific                Asymptotics: T,N -> Infinity
Panel means:  Included                                       sequentially
Time trend:   Not included

ADF regressions: No lags included
------------------------------------------------------------------------------
                                              Fixed-N exact critical values
                    Statistic      p-value         1%      5%      10%
------------------------------------------------------------------------------
 t-bar               -0.9341                    -2.400  -2.150  -2.010
 t-tilde-bar         -0.9346
 Z-t-tilde-bar        1.3796       0.9161
------------------------------------------------------------------------------


Levin-Lin-Chu unit-root test for lnTRPM
---------------------------------------
Ho: Panels contain unit roots               Number of panels  =      4
Ha: Panels are stationary                   Number of periods =    120

AR parameter: Common                        Asymptotics: N/T -> 0
Panel means:  Included
Time trend:   Included

ADF regressions: 1 lag
LR variance:     Bartlett kernel, 15.00 lags average (chosen by LLC)
------------------------------------------------------------------------------
                    Statistic      p-value
------------------------------------------------------------------------------
 Unadjusted t        -3.3774
 Adjusted t*          0.9836       0.8373
------------------------------------------------------------------------------


Im-Pesaran-Shin unit-root test for lnTRPM
-----------------------------------------
Ho: All panels contain unit roots           Number of panels  =      4
Ha: Some panels are stationary              Number of periods =    120

AR parameter: Panel-specific                Asymptotics: T,N -> Infinity
Panel means:  Included                                       sequentially
Time trend:   Included

ADF regressions: No lags included
------------------------------------------------------------------------------
                                              Fixed-N exact critical values
                    Statistic      p-value         1%      5%      10%
------------------------------------------------------------------------------
 t-bar               -0.9685                    -2.990  -2.750  -2.620
 t-tilde-bar         -0.9709
 Z-t-tilde-bar        1.2918       0.9018
------------------------------------------------------------------------------


Levin-Lin-Chu unit-root test for lnPENUSD
-----------------------------------------
Ho: Panels contain unit roots               Number of panels  =      4
Ha: Panels are stationary                   Number of periods =    120

AR parameter: Common                        Asymptotics: N/T -> 0
Panel means:  Included
Time trend:   Included

ADF regressions: 1 lag
LR variance:     Bartlett kernel, 15.00 lags average (chosen by LLC)
------------------------------------------------------------------------------
                    Statistic      p-value
```

```
------------------------------------------------------------------------------
 Unadjusted t        -4.1945
 Adjusted t*          0.5310       0.7023
------------------------------------------------------------------------------


Im-Pesaran-Shin unit-root test for lnPENUSD
-------------------------------------------
Ho: All panels contain unit roots           Number of panels  =      4
Ha: Some panels are stationary              Number of periods =    120

AR parameter: Panel-specific                Asymptotics: T,N -> Infinity
Panel means:  Included                                       sequentially
Time trend:   Included

ADF regressions: No lags included
------------------------------------------------------------------------------
                                              Fixed-N exact critical values
                    Statistic      p-value         1%      5%      10%
------------------------------------------------------------------------------
 t-bar               -2.0098                    -2.990  -2.750  -2.620
 t-tilde-bar         -1.9842
 Z-t-tilde-bar       -1.1585       0.1233
------------------------------------------------------------------------------


Levin-Lin-Chu unit-root test for DlnC
-------------------------------------
Ho: Panels contain unit roots               Number of panels  =      4
Ha: Panels are stationary                   Number of periods =    119

AR parameter: Common                        Asymptotics: N/T -> 0
Panel means:  Included
Time trend:   Included

ADF regressions: 1 lag
LR variance:     Bartlett kernel, 15.00 lags average (chosen by LLC)
------------------------------------------------------------------------------
                    Statistic      p-value
------------------------------------------------------------------------------
 Unadjusted t       -13.6092
 Adjusted t*        -11.9784       0.0000
------------------------------------------------------------------------------


Im-Pesaran-Shin unit-root test for DlnC
---------------------------------------
Ho: All panels contain unit roots           Number of panels  =      4
Ha: Some panels are stationary              Number of periods =    119

AR parameter: Panel-specific                Asymptotics: T,N -> Infinity
Panel means:  Included                                       sequentially
Time trend:   Included

ADF regressions: No lags included
------------------------------------------------------------------------------
                                              Fixed-N exact critical values
                    Statistic      p-value         1%      5%      10%
------------------------------------------------------------------------------
 t-bar               -7.5575                    -2.990  -2.750  -2.620
 t-tilde-bar         -6.2289
 Z-t-tilde-bar      -11.4235       0.0000
------------------------------------------------------------------------------


Levin-Lin-Chu unit-root test for DlnSPPC
----------------------------------------
Ho: Panels contain unit roots               Number of panels  =      4
```

```
Ha: Panels are stationary                 Number of periods =    119

AR parameter: Common                      Asymptotics: N/T -> 0
Panel means:  Included
Time trend:   Included

ADF regressions: 1 lag
LR variance:     Bartlett kernel, 15.00 lags average (chosen by LLC)
------------------------------------------------------------------------------
                    Statistic      p-value
------------------------------------------------------------------------------
 Unadjusted t       -14.2177
 Adjusted t*        -12.7286       0.0000
------------------------------------------------------------------------------


Im-Pesaran-Shin unit-root test for DlnSPPC
------------------------------------------
Ho: All panels contain unit roots           Number of panels  =      4
Ha: Some panels are stationary              Number of periods =    119

AR parameter: Panel-specific                Asymptotics: T,N -> Infinity
Panel means:  Included                                        sequentially
Time trend:   Included

ADF regressions: No lags included
------------------------------------------------------------------------------
                                              Fixed-N exact critical values
                    Statistic      p-value         1%      5%      10%
------------------------------------------------------------------------------
 t-bar               -7.6374                    -2.990  -2.750  -2.620
 t-tilde-bar         -6.2743
 Z-t-tilde-bar      -11.5334       0.0000
------------------------------------------------------------------------------


Levin-Lin-Chu unit-root test for DlnTRPM
----------------------------------------
Ho: Panels contain unit roots               Number of panels  =      4
Ha: Panels are stationary                   Number of periods =    119

AR parameter: Common                        Asymptotics: N/T -> 0
Panel means:  Included
Time trend:   Not included

ADF regressions: 1 lag
LR variance:     Bartlett kernel, 15.00 lags average (chosen by LLC)
------------------------------------------------------------------------------
                    Statistic      p-value
------------------------------------------------------------------------------
 Unadjusted t       -11.7343
 Adjusted t*         -8.5435       0.0000
------------------------------------------------------------------------------


Im-Pesaran-Shin unit-root test for DlnTRPM
------------------------------------------
Ho: All panels contain unit roots           Number of panels  =      4
Ha: Some panels are stationary              Number of periods =    119

AR parameter: Panel-specific                Asymptotics: T,N -> Infinity
Panel means:  Included                                        sequentially
Time trend:   Not included

ADF regressions: No lags included
------------------------------------------------------------------------------
                                              Fixed-N exact critical values
                    Statistic      p-value         1%      5%      10%
------------------------------------------------------------------------------
 t-bar               -6.4976                    -2.400  -2.150  -2.010
 t-tilde-bar         -5.5875
```

```
 Z-t-tilde-bar      -9.8723       0.0000
------------------------------------------------------------------------------


Levin-Lin-Chu unit-root test for DlnPENUSD
------------------------------------------
Ho: Panels contain unit roots             Number of panels  =      4
Ha: Panels are stationary                 Number of periods =    119

AR parameter: Common                      Asymptotics: N/T -> 0
Panel means:  Included
Time trend:   Included

ADF regressions: 1 lag
LR variance:     Bartlett kernel, 15.00 lags average (chosen by LLC)
------------------------------------------------------------------------------
                    Statistic      p-value
------------------------------------------------------------------------------
 Unadjusted t       -14.8270
 Adjusted t*        -12.7260       0.0000
------------------------------------------------------------------------------


Im-Pesaran-Shin unit-root test for DlnPENUSD
--------------------------------------------
Ho: All panels contain unit roots         Number of panels  =      4
Ha: Some panels are stationary            Number of periods =    119

AR parameter: Panel-specific              Asymptotics: T,N -> Infinity
Panel means:  Included                                  sequentially
Time trend:   Included

ADF regressions: No lags included
------------------------------------------------------------------------------
                                            Fixed-N exact critical values
                    Statistic      p-value         1%       5%      10%
------------------------------------------------------------------------------
 t-bar               -9.9608                     -2.990  -2.750  -2.620
 t-tilde-bar         -7.3612
 Z-t-tilde-bar      -14.1618       0.0000
------------------------------------------------------------------------------


Levin-Lin-Chu unit-root test for DlnPENUSD
------------------------------------------
Ho: Panels contain unit roots             Number of panels  =      4
Ha: Panels are stationary                 Number of periods =    119

AR parameter: Common                      Asymptotics: N/T -> 0
Panel means:  Included
Time trend:   Not included

ADF regressions: 1 lag
LR variance:     Bartlett kernel, 15.00 lags average (chosen by LLC)
------------------------------------------------------------------------------
                    Statistic      p-value
------------------------------------------------------------------------------
 Unadjusted t       -14.7275
 Adjusted t*        -11.8032       0.0000
------------------------------------------------------------------------------


Im-Pesaran-Shin unit-root test for DlnPENUSD
--------------------------------------------
Ho: All panels contain unit roots         Number of panels  =      4
Ha: Some panels are stationary            Number of periods =    119

AR parameter: Panel-specific              Asymptotics: T,N -> Infinity
Panel means:  Included                                  sequentially
Time trend:   Not included
```

ADF regressions: No lags included

| | Statistic | p-value | Fixed-N exact critical values 1% | 5% | 10% |
|---|---|---|---|---|---|
| t-bar | -9.9588 | | -2.400 | -2.150 | -2.010 |
| t-tilde-bar | -7.3435 | | | | |
| Z-t-tilde-bar | -14.1188 | 0.0000 | | | |

.

# Prueba de Multicolinealidad: Correlograma, VIF y PCA

| | DlnC | PBI | DlnSPPC | DlnTRPM | IGBVL | DlnPEN~D | INF | RETIRO |
|---|---|---|---|---|---|---|---|---|
| DlnC | 1.0000 | | | | | | | |
| PBI | -0.1488 | 1.0000 | | | | | | |
| DlnSPPC | 0.8492 | -0.0757 | 1.0000 | | | | | |
| DlnTRPM | -0.0677 | 0.4205 | -0.0699 | 1.0000 | | | | |
| IGBVL | -0.1672 | 0.6441 | -0.0484 | 0.0603 | 1.0000 | | | |
| DlnPENUSD | -0.1874 | -0.0489 | -0.2594 | 0.1081 | -0.1819 | 1.0000 | | |
| INF | -0.1817 | 0.0997 | -0.1929 | 0.2550 | -0.0711 | 0.0537 | 1.0000 | |
| RETIRO | 0.0447 | -0.0380 | 0.0851 | -0.0675 | 0.0941 | 0.0716 | 0.0413 | 1.0000 |

| Source | SS | df | MS | | |
|---|---|---|---|---|---|
| | | | | Number of obs = | 476 |
| | | | | F(8, 467) = | 1239.37 |
| Model | 5.9716e+12 | 8 | 7.4645e+11 | Prob > F = | 0.0000 |
| Residual | 2.8126e+11 | 467 | 602278449 | R-squared = | 0.9550 |
| | | | | Adj R-squared = | 0.9542 |
| Total | 6.2528e+12 | 475 | 1.3164e+10 | Root MSE = | 24541 |

| M | Coef. | Std. Err. | t | P>\|t\| | [95% Conf. | Interval] |
|---|---|---|---|---|---|---|
| DlnC | -82507.3 | 65367.48 | -1.26 | 0.208 | -210958.1 | 45943.52 |
| PBI | -.1374263 | .0353338 | -3.89 | 0.000 | -.2068592 | -.0679933 |
| DlnSPPC | -144664.7 | 80898.71 | -1.79 | 0.074 | -303635.2 | 14305.88 |
| DlnTRPM | 25080.72 | 6921.552 | 3.62 | 0.000 | 11479.48 | 38681.97 |
| IGBVL | 23.9689 | .3411795 | 70.25 | 0.000 | 23.29846 | 24.63934 |
| DlnPENUSD | 47927.07 | 83858.05 | 0.57 | 0.568 | -116858.8 | 212712.9 |
| INF | -25637.08 | 6800.735 | -3.77 | 0.000 | -39000.91 | -12273.25 |
| RETIRO | .987728 | .2488422 | 3.97 | 0.000 | .498739 | 1.476717 |
| _cons | 132453.7 | 15357.86 | 8.62 | 0.000 | 102274.6 | 162632.8 |

| Variable | VIF | 1/VIF |
|---|---|---|
| DlnSPPC | 3.85 | 0.259547 |
| DlnC | 3.82 | 0.261719 |
| PBI | 2.33 | 0.430085 |
| IGBVL | 2.09 | 0.477895 |
| DlnTRPM | 1.40 | 0.712797 |
| DlnPENUSD | 1.16 | 0.865172 |
| INF | 1.14 | 0.877486 |
| RETIRO | 1.06 | 0.939963 |
| Mean VIF | 2.11 | |

## Prueba de endogeneidad o Durbin-Wu-Hausman

```
Fixed-effects (within) regression               Number of obs      =       476
Group variable: AFP                             Number of groups   =         4

R-sq:                                           Obs per group:
     within  = 0.9378                                          min =       119
     between =      .                                          avg =     119.0
     overall = 0.9344                                          max =       119

                                                F(7,465)           =   1001.27
corr(u_i, Xb)  = -0.0597                        Prob > F           =    0.0000

------------------------------------------------------------------------------
         lnM |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        DlnC |  -1.021763   .2325592    -4.39   0.000     -1.47876   -.5647655
         PBI |  -2.13e-07   8.28e-08    -2.58   0.010    -3.76e-07   -5.06e-08
     DlnSPPC |   .6515438   .2601173     2.50   0.013     .1403928    1.162695
     DlnTRPM |   .0436178   .0161822     2.70   0.007     .0118186    .0754171
       IGBVL |   .0000462   8.01e-07    57.67   0.000     .0000446    .0000478
         INF |  -.0402038   .0159672    -2.52   0.012    -.0715806   -.008827
      RETIRO |   1.67e-06   5.79e-07     2.88   0.004     5.31e-07    2.81e-06
       _cons |   12.36561   .0363326   340.35   0.000     12.29422    12.43701
-------------+----------------------------------------------------------------
     sigma_u |  .01525904
     sigma_e |  .05760389
         rho |  .06556894   (fraction of variance due to u_i)
------------------------------------------------------------------------------
F test that all u_i=0: F(3, 465) = 3.63                      Prob > F = 0.0129


Random-effects GLS regression                   Number of obs      =       476
Group variable: AFP                             Number of groups   =         4

R-sq:                                           Obs per group:
     within  = 0.0000                                          min =       119
     between = 0.0000                                          avg =     119.0
     overall = 0.9363                                          max =       119

                                                Wald chi2(7)       =   6881.84
corr(u_i, X)   = 0 (assumed)                    Prob > chi2        =    0.0000

------------------------------------------------------------------------------
         lnM |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        DlnC |  -.4446969   .1546932    -2.87   0.004      -.74789   -.1415038
         PBI |  -2.11e-07   8.35e-08    -2.52   0.012    -3.75e-07   -4.70e-08
     DlnSPPC |   .0549713   .1886812     0.29   0.771    -.3148371    .4247796
     DlnTRPM |   .0428349   .0163164     2.63   0.009     .0108552    .0748145
       IGBVL |   .0000467   7.93e-07    58.89   0.000     .0000451    .0000482
         INF |  -.0369986   .0160716    -2.30   0.021    -.0684984   -.0054988
      RETIRO |   1.72e-06   5.83e-07     2.96   0.003     5.82e-07    2.87e-06
       _cons |   12.34999   .0363258   339.98   0.000     12.27879    12.42118
-------------+----------------------------------------------------------------
     sigma_u |          0
     sigma_e |  .05760389
         rho |          0   (fraction of variance due to u_i)
------------------------------------------------------------------------------


Note: the rank of the differenced variance matrix (4) does not equal the number of
coefficients being tested (7); be sure this is what you expect, or there may be problems
computing the test.  Examine the output of your estimators for anything unexpected and
possibly consider scaling your variables so that the coefficients are on a similar
scale.
```

```
                 ---- Coefficients ----
             |      (b)          (B)            (b-B)     sqrt(diag(V_b-V_B))
             |     fixed        random       Difference          S.E.
-------------+----------------------------------------------------------------
        DlnC |   -1.021763    -.4446969       -.5770658        .1736485
         PBI |   -2.13e-07    -2.11e-07       -2.64e-09               .
     DlnSPPC |    .6515438     .0549713        .5965725        .1790543
     DlnTRPM |    .0436178     .0428349        .0007829               .
       IGBVL |    .0000462     .0000467       -5.03e-07        1.13e-07
         INF |   -.0402038    -.0369986       -.0032052               .
      RETIRO |    1.67e-06     1.72e-06       -5.66e-08               .
------------------------------------------------------------------------------
                           b = consistent under Ho and Ha; obtained from xtreg
            B = inconsistent under Ha, efficient under Ho; obtained from xtreg

    Test:  Ho:  difference in coefficients not systematic

                  chi2(4) = (b-B)'[(V_b-V_B)^(-1)](b-B)
                          =       11.04
                Prob>chi2 =      0.0261
                (V_b-V_B is not positive definite)


Fixed-effects (within) regression               Number of obs      =       476
Group variable: AFP                             Number of groups   =         4

R-sq:                                           Obs per group:
     within  = 0.9368                                         min =        119
     between =      .                                         avg =      119.0
     overall = 0.9334                                         max =        119

                                                F(6,466)           =   1151.91
corr(u_i, Xb)  = -0.0610                        Prob > F           =    0.0000

------------------------------------------------------------------------------
         lnM |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        DlnC |  -1.043512   .2338847    -4.46   0.000    -1.503111   -.5839126
       lnPBI |  -.1324653   .0427306    -3.10   0.002    -.2164337   -.0484969
     DlnSPPC |   .7042845   .2610865     2.70   0.007      .191232    1.217337
     DlnTRPM |   .0484796   .0170091     2.85   0.005     .0150555    .0819037
       IGBVL |   .0000466   7.90e-07    58.95   0.000      .000045    .0000481
         INF |  -.0369935   .0159805    -2.31   0.021    -.0683963   -.0055907
       _cons |   13.99125   .5533719    25.28   0.000     12.90384    15.07867
-------------+----------------------------------------------------------------
     sigma_u |  .01558419
     sigma_e |  .05797913
         rho |  .06737982   (fraction of variance due to u_i)
------------------------------------------------------------------------------
F test that all u_i=0: F(3, 466) = 3.74                      Prob > F = 0.0111


Random-effects GLS regression                   Number of obs      =       476
Group variable: AFP                             Number of groups   =         4

R-sq:                                           Obs per group:
     within  = 0.0000                                         min =        119
     between = 0.0000                                         avg =      119.0
     overall = 0.9353                                         max =        119

                                                Wald chi2(6)       =   6781.20
corr(u_i, X)   = 0 (assumed)                    Prob > chi2        =    0.0000

------------------------------------------------------------------------------
         lnM |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        DlnC |  -.4545649   .1556978    -2.92   0.004    -.7597269   -.1494029
       lnPBI |  -.1322999   .0431039    -3.07   0.002    -.2167821   -.0478177
     DlnSPPC |   .0963878   .1894406     0.51   0.611     -.274909    .4676846
     DlnTRPM |   .0477701   .0171564     2.78   0.005     .0141441    .0813961
```

```
       IGBVL |   .0000471   7.81e-07    60.35   0.000     .0000456    .0000486
         INF |  -.0335493   .0160868    -2.09   0.037    -.0650789   -.0020197
       _cons |   13.97416   .5581841    25.04   0.000     12.88014    15.06818
-------------+----------------------------------------------------------------
     sigma_u |          0
     sigma_e |  .05797913
         rho |          0   (fraction of variance due to u_i)
------------------------------------------------------------------------------


Note: the rank of the differenced variance matrix (5) does not equal the number of
coefficients
        being tested (6); be sure this is what you expect, or there may be problems
computing
        the test.  Examine the output of your estimators for anything unexpected and
possibly
        consider scaling your variables so that the coefficients are on a similar scale.

                 ---- Coefficients ----
             |      (b)          (B)            (b-B)     sqrt(diag(V_b-V_B))
             |     fixed        random       Difference          S.E.
-------------+----------------------------------------------------------------
        DlnC |   -1.043512    -.4545649       -.5889469        .1745286
       lnPBI |   -.1324653    -.1322999       -.0001654               .
     DlnSPPC |    .7042845     .0963878        .6078967        .1796618
     DlnTRPM |    .0484796     .0477701        .0007095               .
       IGBVL |    .0000466     .0000471       -5.35e-07        1.22e-07
         INF |   -.0369935    -.0335493       -.0034442               .
------------------------------------------------------------------------------
                           b = consistent under Ho and Ha; obtained from xtreg
            B = inconsistent under Ha, efficient under Ho; obtained from xtreg

    Test:  Ho:  difference in coefficients not systematic

                  chi2(5) = (b-B)'[(V_b-V_B)^(-1)](b-B)
                          =       11.39
                Prob>chi2 =      0.0442
                (V_b-V_B is not positive definite)
```

# Prueba de Multiplicadores de Breusch y Pagan Lagrangiano para Efectos Aleatorios

```
Random-effects GLS regression                   Number of obs     =        476
Group variable: AFP                             Number of groups  =          4

R-sq:                                           Obs per group:
     within  = 0.0000                                         min =        119
     between = 0.0000                                         avg =      119.0
     overall = 0.9353                                         max =        119

                                                Wald chi2(6)      =    6781.20
corr(u_i, X)   = 0 (assumed)                    Prob > chi2       =     0.0000

------------------------------------------------------------------------------
         lnM |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        DlnC |  -.4545649   .1556978    -2.92   0.004    -.7597269   -.1494029
       lnPBI |  -.1322999   .0431039    -3.07   0.002    -.2167821   -.0478177
     DlnSPPC |   .0963878   .1894406     0.51   0.611     -.274909    .4676846
     DlnTRPM |   .0477701   .0171564     2.78   0.005     .0141441    .0813961
       IGBVL |   .0000471   7.81e-07    60.35   0.000     .0000456    .0000486
         INF |  -.0335493   .0160868    -2.09   0.037    -.0650789   -.0020197
       _cons |   13.97416   .5581841    25.04   0.000     12.88014    15.06818
-------------+----------------------------------------------------------------
     sigma_u |          0
     sigma_e |  .05797913
         rho |          0   (fraction of variance due to u_i)
------------------------------------------------------------------------------


Breusch and Pagan Lagrangian multiplier test for random effects

        lnM[AFP,t] = Xb + u[AFP] + e[AFP,t]

        Estimated results:
                         |       Var     sd = sqrt(Var)
                ---------+-----------------------------
                     lnM |   .0522104       .2284961
                       e |   .0033616       .0579791
                       u |          0              0

        Test:   Var(u) = 0
                             chibar2(01) =     0.00
                          Prob > chibar2 =   1.0000
```

# Prueba de Independencia de Breusch-Pagan LM prueba para Efectos Fijos

```
Fixed-effects (within) regression               Number of obs     =        476
Group variable: AFP                             Number of groups  =          4

R-sq:                                           Obs per group:
     within  = 0.9368                                         min =        119
     between =      .                                         avg =      119.0
     overall = 0.9334                                         max =        119

                                                F(6,466)          =    1151.91
corr(u_i, Xb)  = -0.0610                        Prob > F          =     0.0000

------------------------------------------------------------------------------
         lnM |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        DlnC |  -1.043512   .2338847    -4.46   0.000    -1.503111   -.5839126
       lnPBI |  -.1324653   .0427306    -3.10   0.002    -.2164337   -.0484969
     DlnSPPC |   .7042845   .2610865     2.70   0.007      .191232    1.217337
     DlnTRPM |   .0484796   .0170091     2.85   0.005     .0150555    .0819037
       IGBVL |   .0000466   7.90e-07    58.95   0.000      .000045    .0000481
         INF |  -.0369935   .0159805    -2.31   0.021    -.0683963   -.0055907
       _cons |   13.99125   .5533719    25.28   0.000     12.90384    15.07867
-------------+----------------------------------------------------------------
     sigma_u |  .01558419
     sigma_e |  .05797913
         rho |  .06737982   (fraction of variance due to u_i)
------------------------------------------------------------------------------
F test that all u_i=0: F(3, 466) = 3.74                      Prob > F = 0.0111


Correlation matrix of residuals:

             |      __e1       __e2       __e3       __e4
-------------+--------------------------------------------
        __e1 |  .3328731
        __e2 |  .3375114   .4168475
        __e3 |  .3312955   .4089507   .4069753
        __e4 |  .3381331   .4117539   .4049454   .4098001

        __e1    __e2    __e3    __e4
__e1  1.0000
__e2  0.9061  1.0000
__e3  0.9001  0.9929  1.0000
__e4  0.9155  0.9962  0.9916  1.0000

Breusch-Pagan LM test of independence: chi2(6) =   646.269, Pr = 0.0000
Based on 119 complete observations over panel units
```

## Prueba de Heterocedasticidad grupal de Wald modificada en un modelo de efectos fijos

```
Modified Wald test for groupwise heteroskedasticity
in fixed effect regression model

H0: sigma(i)^2 = sigma^2 for all i

chi2 (4)  =        2.23
Prob>chi2 =      0.6929
```

```
. xtreg lnM DlnC lnPBI DlnSPPC DlnTRPM IGBVL INF, fe robust

Fixed-effects (within) regression               Number of obs     =        476
Group variable: AFP                             Number of groups  =          4

R-sq:                                           Obs per group:
     within  = 0.9368                                         min =        119
     between =      .                                         avg =      119.0
     overall = 0.9334                                         max =        119

                                                F(4,3)            =          .
corr(u_i, Xb)  = -0.0610                        Prob > F          =          .

                                   (Std. Err. adjusted for 4 clusters in AFP)
------------------------------------------------------------------------------
             |               Robust
         lnM |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        DlnC |  -1.043512   .2321915    -4.49   0.021    -1.782449   -.3045749
       lnPBI |  -.1324653   .0017615   -75.20   0.000    -.1380712   -.1268594
     DlnSPPC |   .7042845   .2603881     2.70   0.073    -.1243865    1.532956
     DlnTRPM |   .0484796   .0024326    19.93   0.000     .0407379    .0562213
       IGBVL |   .0000466   1.10e-06    42.30   0.000     .0000431    .0000501
         INF |  -.0369935   .0083871    -4.41   0.022    -.0636851   -.0103019
       _cons |   13.99125   .0396669   352.72   0.000     13.86502    14.11749
-------------+----------------------------------------------------------------
     sigma_u |  .01558419
     sigma_e |  .05797913
         rho |  .06737982   (fraction of variance due to u_i)
------------------------------------------------------------------------------
```

## Prueba de Heterocedasticidad de Breusch-Pagan a partir de los residuos para efectos fijos

```
Fixed-effects (within) regression               Number of obs     =        476
Group variable: AFP                             Number of groups  =          4

R-sq:                                           Obs per group:
     within  = 0.9368                                         min =        119
     between =      .                                         avg =      119.0
     overall = 0.9334                                         max =        119

                                                F(6,466)          =    1151.91
corr(u_i, Xb)  = -0.0610                        Prob > F          =     0.0000

------------------------------------------------------------------------------
         lnM |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        DlnC |  -1.043512   .2338847    -4.46   0.000    -1.503111   -.5839126
       lnPBI |  -.1324653   .0427306    -3.10   0.002    -.2164337   -.0484969
     DlnSPPC |   .7042845   .2610865     2.70   0.007      .191232    1.217337
     DlnTRPM |   .0484796   .0170091     2.85   0.005     .0150555    .0819037
       IGBVL |   .0000466   7.90e-07    58.95   0.000      .000045    .0000481
         INF |  -.0369935   .0159805    -2.31   0.021    -.0683963   -.0055907
       _cons |   13.99125   .5533719    25.28   0.000     12.90384    15.07867
-------------+----------------------------------------------------------------
     sigma_u |  .01558419
     sigma_e |  .05797913
         rho |  .06737982   (fraction of variance due to u_i)
------------------------------------------------------------------------------
F test that all u_i=0: F(3, 466) = 3.74                      Prob > F = 0.0111


      Source |       SS           df       MS      Number of obs   =       476
-------------+----------------------------------   F(7, 468)       =      5.82
       Model |  .000683525         7  .000097646   Prob > F        =    0.0000
    Residual |   .00784595       468  .000016765   R-squared       =    0.0801
-------------+----------------------------------   Adj R-squared   =    0.0664
       Total |  .008529475       475  .000017957   Root MSE        =    .00409

------------------------------------------------------------------------------
          e2 |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
         lnM |   .0153464   .0032327     4.75   0.000      .008994    .0216987
        DlnC |  -.0155487   .0109987    -1.41   0.158    -.0371617    .0060643
       lnPBI |    .004645   .0030478     1.52   0.128    -.0013441     .010634
     DlnSPPC |  -.0012982   .0132661    -0.10   0.922    -.0273666    .0247703
     DlnTRPM |  -.0013504    .001211    -1.12   0.265    -.0037301    .0010292
       IGBVL |  -7.24e-07   1.62e-07    -4.48   0.000    -1.04e-06   -4.06e-07
         INF |   .0002126   .0011314     0.19   0.851    -.0020107    .0024359
       _cons |  -.2454037   .0597306    -4.11   0.000    -.3627771   -.1280304
------------------------------------------------------------------------------
```

# Prueba de Autocorrelación

```
Linear regression                               Number of obs     =        472
                                                F(3, 3)           =          .
                                                Prob > F          =          .
                                                R-squared         =     0.7104
                                                Root MSE          =      .0263

                                    (Std. Err. adjusted for 4 clusters in AFP)
------------------------------------------------------------------------------
             |               Robust
       D.lnM |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        DlnC |
         D1. |   -.065544   .0659801    -0.99   0.394    -.2755221    .1444341
             |
       lnPBI |
         D1. |   -.029465   .0003057   -96.40   0.000    -.0304377   -.0284922
             |
     DlnSPPC |
         D1. |   .0402934   .0662844     0.61   0.586    -.1706531    .2512399
             |
     DlnTRPM |
         D1. |   .0054437   .0000596    91.34   0.000     .0052541    .0056334
             |
       IGBVL |
         D1. |   .0000375   2.28e-08  1644.16   0.000     .0000374    .0000376
             |
         INF |
         D1. |  -.0271344   .0001868  -145.25   0.000     -.027729   -.0265399
------------------------------------------------------------------------------

Wooldridge test for autocorrelation in panel data
H0: no first-order autocorrelation
    F(  1,       3) = 104354.895
           Prob > F =      0.0000


FE (within) regression with AR(1) disturbances  Number of obs     =        472
Group variable: AFP                             Number of groups  =          4

R-sq:                                           Obs per group:
     within  = 0.7290                                         min =        118
     between =      .                                         avg =      118.0
     overall = 0.9316                                         max =        118

                                                F(6,462)          =     207.18
corr(u_i, Xb)  = -0.0050                        Prob > F          =     0.0000

------------------------------------------------------------------------------
         lnM |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        DlnC |   -.070915   .1756261    -0.40   0.687    -.4160398    .2742099
       lnPBI |  -.0329679   .0192525    -1.71   0.087    -.0708012    .0048653
     DlnSPPC |   .0403662   .1832587     0.22   0.826    -.3197578    .4004901
     DlnTRPM |   .0058039   .0067532     0.86   0.391    -.0074669    .0190748
       IGBVL |   .0000378   1.26e-06    30.03   0.000     .0000353    .0000403
         INF |  -.0262252   .0087968    -2.98   0.003    -.0435119   -.0089386
       _cons |   12.85268   .0179143   717.45   0.000     12.81748    12.88789
-------------+----------------------------------------------------------------
      rho_ar |  .92928696
     sigma_u |  .00103941
     sigma_e |  .02579806
     rho_fov |  .00162067   (fraction of variance because of u_i)
------------------------------------------------------------------------------
F test that all u_i=0: F(3,462) = 0.00                       Prob > F = 1.0000
```

# Prueba de Normalidad

```
Fixed-effects (within) regression               Number of obs      =       476
Group variable: AFP                             Number of groups   =         4

R-sq:                                           Obs per group:
     within  = 0.9368                                         min =       119
     between =      .                                         avg =     119.0
     overall = 0.9334                                         max =       119

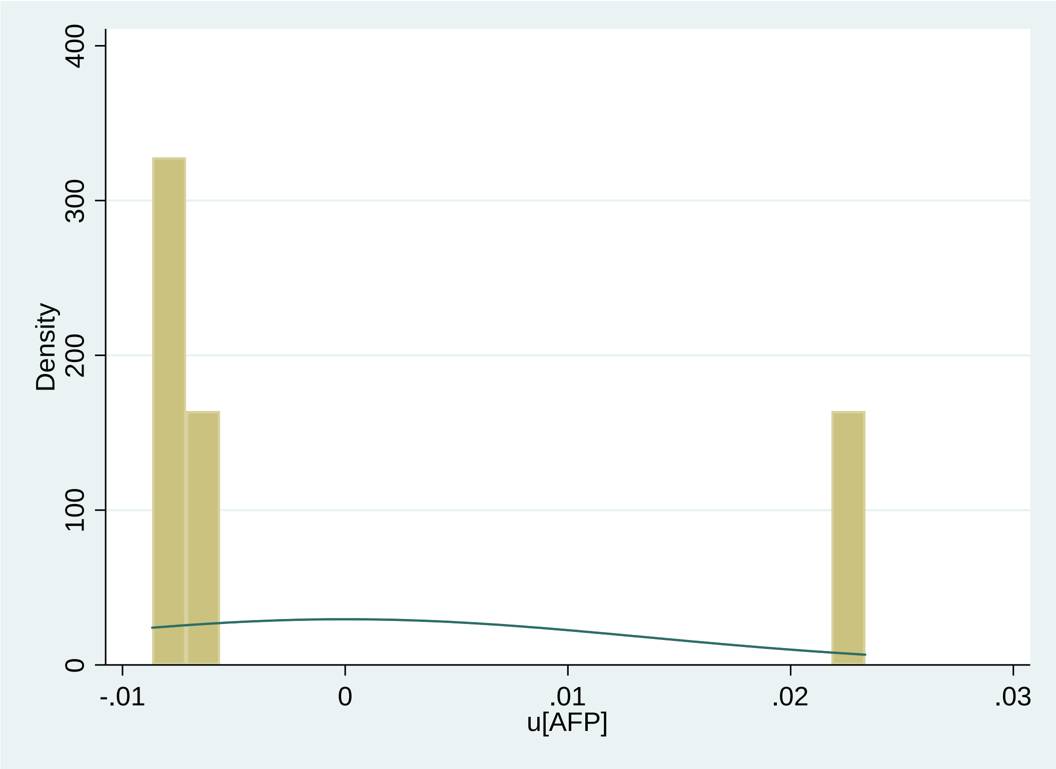
                                                F(6,466)           =   1151.91
corr(u_i, Xb)  = -0.0610                        Prob > F           =    0.0000

------------------------------------------------------------------------------
         lnM |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        DlnC |  -1.043512   .2338847    -4.46   0.000    -1.503111   -.5839126
       lnPBI |  -.1324653   .0427306    -3.10   0.002    -.2164337   -.0484969
     DlnSPPC |   .7042845   .2610865     2.70   0.007      .191232    1.217337
     DlnTRPM |   .0484796   .0170091     2.85   0.005     .0150555    .0819037
       IGBVL |   .0000466   7.90e-07    58.95   0.000      .000045    .0000481
         INF |  -.0369935   .0159805    -2.31   0.021    -.0683963   -.0055907
       _cons |   13.99125   .5533719    25.28   0.000     12.90384    15.07867
-------------+----------------------------------------------------------------
     sigma_u |  .01558419
     sigma_e |  .05797913
         rho |  .06737982   (fraction of variance due to u_i)
------------------------------------------------------------------------------
F test that all u_i=0: F(3, 466) = 3.74                      Prob > F = 0.0111

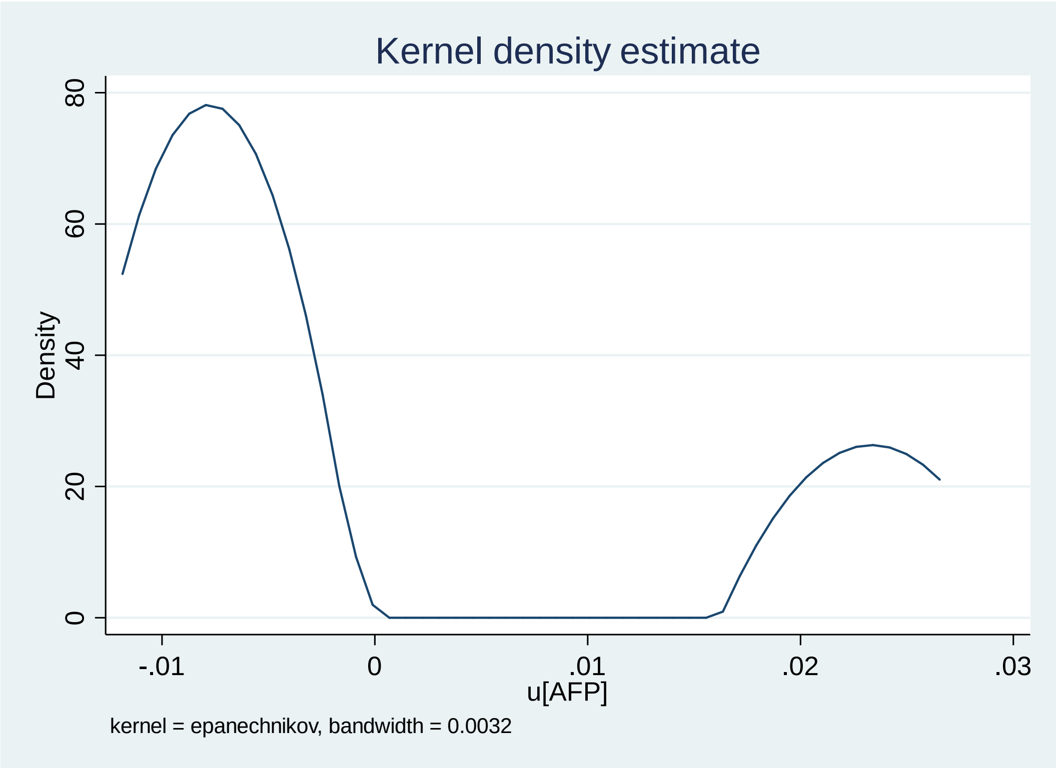
    Variable |        Obs        Mean    Std. Dev.       Min        Max
-------------+---------------------------------------------------------
   residuals |        476           0    .0135105  -.0086697   .0233527

    Variable |        Obs        Mean    Std. Dev.       Min        Max
-------------+---------------------------------------------------------
        skew |        476    1.144092    2.323855  -.2642385   5.164125
        kurt |        476    2.318225    3.819245   .0703702   8.926113

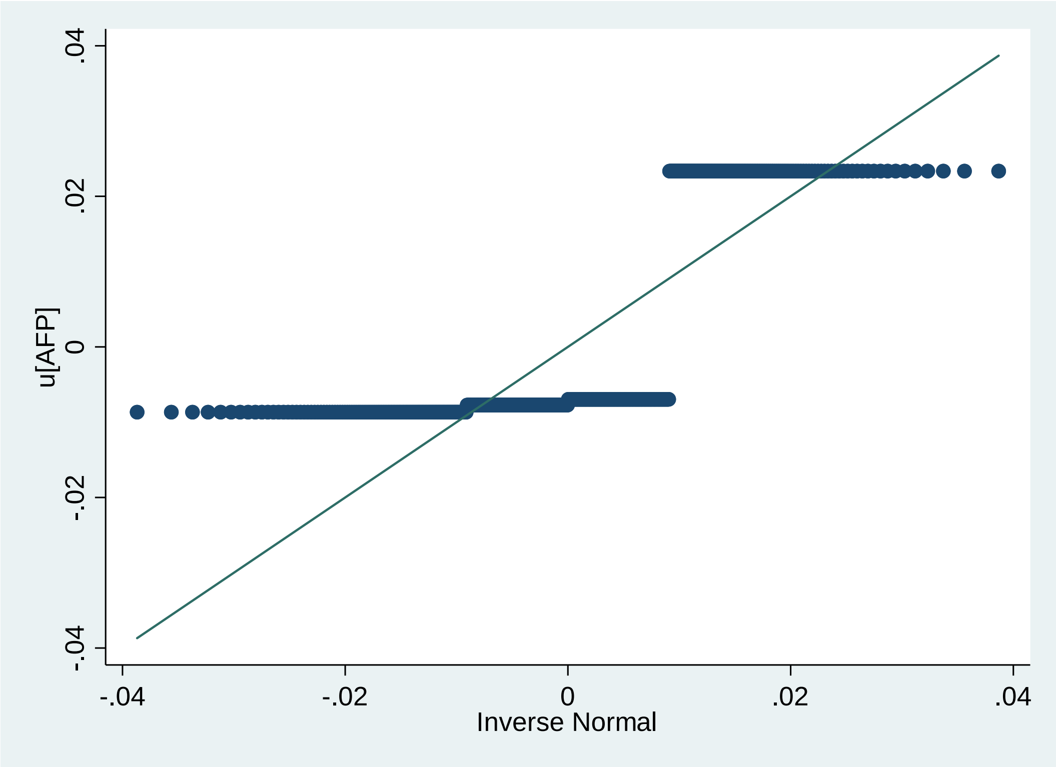
(bin=21, start=-.00866969, width=.00152488)
```



Density
400
300
200
100
0
-.01
0
.01
.02
.03
u[AFP]

Kernel density estimate
Density
80
60
40
20
0
-.01
0
.01
.02
.03
u[AFP]
kernel = epanechnikov, bandwidth = 0.0032




```
                    Skewness/Kurtosis tests for Normality
                                                          ------ joint ------
    Variable |        Obs   Pr(Skewness)   Pr(Kurtosis) adj chi2(2)    Prob>chi2
-------------+---------------------------------------------------------------
   residuals |        476      0.0000         0.0000       71.28         0.0000

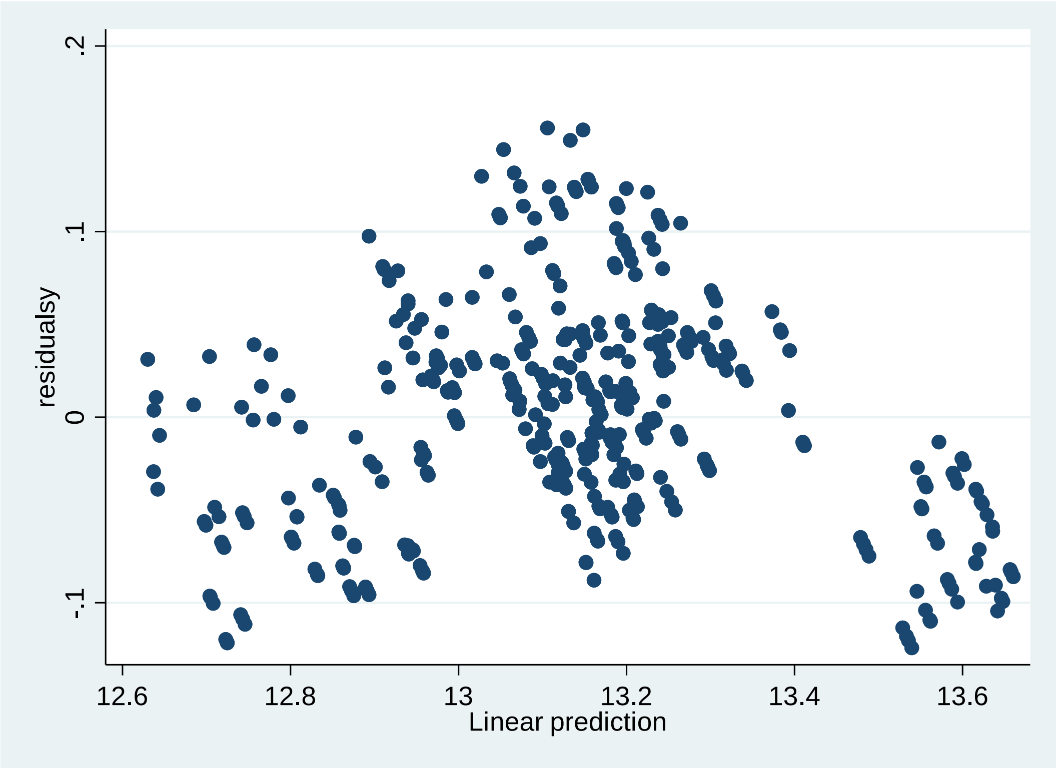
    Variable |        Obs        Mean    Std. Dev.       Min        Max
-------------+---------------------------------------------------------
        skew |        476    1.144092    2.323855  -.2642385   5.164125

    Variable |        Obs        Mean    Std. Dev.       Min        Max
-------------+---------------------------------------------------------
        kurt |        476    2.318225    3.819245   .0703702   8.926113

. display jb
.73506533

. display p_value
.69244071
```

Prueba de Homocedasticidad



# Prueba de Estabilidad

```
    Variable |        Obs        Mean    Std. Dev.       Min        Max
-------------+---------------------------------------------------------
cum_residual |        480    1.417606    .7917397          0   2.778971


                        cum_residual
-------------------------------------------------------------
      Percentiles      Smallest
 1%     .0233527              0
 5%     .1437505              0
10%     .2991043       .0086697       Obs                 480
25%     .7507747       .0173394       Sum of Wgt.         480

50%      1.43181                      Mean           1.417606
                        Largest       Std. Dev.      .7917397
75%     2.087632       2.763523
90%     2.504753       2.771247       Variance       .6268517
95%     2.643793       2.778971       Skewness      -.0508359
99%     2.755798       2.778971       Kurtosis       1.862901
```

# Prueba de Error en especificación en regresión

```
note: DlnC omitted because of collinearity

      Source |       SS           df       MS      Number of obs   =       476
-------------+----------------------------------   F(8, 467)       =   1370.57
       Model |   23.786838         8  2.97335475   Prob > F        =    0.0000
    Residual |  1.01312562       467  .002169434   R-squared       =    0.9591
-------------+----------------------------------   Adj R-squared   =    0.9584
       Total |  24.7999636       475   .05221045   Root MSE        =    .04658

------------------------------------------------------------------------------
         lnM |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        yhat |  -193.0013   72.55735    -2.66   0.008    -335.5806   -50.42194
     yhat_sq |   15.17236    5.51625     2.75   0.006     4.332617    26.01211
     yhat_cu |  -.3970566   .1397191    -2.84   0.005    -.6716126   -.1225006
        DlnC |          0  (omitted)
       lnPBI |  -.1834088   .0383242    -4.79   0.000     -.258718   -.1080997
     DlnSPPC |  -.5325265   .0938557    -5.67   0.000    -.7169583   -.3480947
     DlnTRPM |   .0427287    .014957     2.86   0.004     .0133375      .07212
       IGBVL |   .0000488   5.94e-06     8.22   0.000     .0000372    .0000605
         INF |  -.0578642   .0136218    -4.25   0.000    -.0846319   -.0310966
       _cons |   831.8459   317.9837     2.62   0.009     206.9899    1456.702
------------------------------------------------------------------------------

Ramsey RESET test using powers of the fitted values of lnM
       Ho:  model has no omitted variables
                 F(3, 464) =      5.58
                  Prob > F =      0.0009
```

# Estimación del intercepto específico para cada grupo

```
     +------------------+
     | AFP            u |
     |------------------|
  1. |   1     .0233527 |
121. |   2    -.0077245 |
241. |   3    -.0069585 |
361. |   4    -.0086697 |
     +------------------+
```